
Pessimistic Off-Policy Learning under Covariate Shift and Local Differential Privacy

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Abstract

We study off-policy learning from observational data under covariate shift and local differential privacy (LDP) constraints. Modern policy learning faces a fundamental trilemma: incorporating high-dimensional contexts improves performance but requires pooling external data, which introduces distribution shifts and privacy risks. We formalize the difficulty of this problem using the LDP parameter, the transfer exponent, and a relaxed, localized treatment coverage assumption rather than strict global overlap. Under this framework, we establish minimax lower bounds on the policy sub-optimal gap for both non-private and ε -LDP settings. To achieve these fundamental limits, we develop a policy learning algorithm grounded in a unified pessimism principle that addresses both covariate shift and privacy-induced uncertainty. By estimating the conditional outcome via nonparametric regression and constructing a lower confidence bound that penalizes underexplored treatments and privacy noise, our approach bypasses the estimation of possibly heterogeneous historical logging policies. We prove that maximizing this pessimistic objective yields a minimax-optimal policy in the non-private setting, fully characterizing the theoretical trade-offs between data dimensionality and distribution shift. We conjecture that this pessimistic policy achieves matching optimality under privacy constraints.

1 Introduction

Personalized decision-making systems use data-driven policies that map personal contexts to optimal treatments. Driven by the rise of fintech and insurtech, the lending and insurance industries readily embrace this approach, utilizing non-traditional, sensitive data such as job histories, email domains, device types, and telematics (Di Maggio and Ratnadiwakara, 2026; McFall et al., 2020). Although explicit personalization is often deemed unfair when groups defined by sensitive traits are differentially rated (Allender et al., 2021; Zhang and Xu, 2024), firms continue personalizing via implicit methods, including targeted rebates (Adida, 2021; Kallus and Zhou, 2018) or customized ad loads (Goli et al., 2025).

Training personalized decision-making systems is cast as a policy learning problem where a distribution P generates n tuples of context, treatment, and potential outcome:

$$(\mathbf{X}_i, T_i, Y_i(\cdot)) \in \mathcal{X} \times \mathcal{T} \times \mathcal{F}, \quad i = 1, \dots, n, \quad (1)$$

where $\mathcal{X} = [0, 1]^d$, $\mathcal{T} = [0, 1]$, and $\mathcal{F} = \{y : \mathcal{T} \rightarrow [0, 1]\}$. Following Rubin’s potential outcomes framework (Rubin, 1984), $Y_i(t)$ represents the outcome that would be obtained if treatment t were applied. However, the learning algorithm only observes the realized outcome corresponding to the chosen treatment:

$$\mathcal{D}_n := \{(\mathbf{X}_i, T_i, Y_i)\}_{i=1}^n, \quad (2)$$

where we use the shorthand $Y_i := Y_i(T_i)$. A policy learning algorithm maps $\mathcal{D}_n$ to a policy $\hat{\pi} : \mathcal{X} \rightarrow \mathcal{T}$, and its central hurdle is optimizing a decision rule without the missing counterfactuals $\{Y_i(t)\}_{t \neq T_i}$. Treatment T_i may represent loan interest rates (Keskin and Zeevi, 2014), surge price multipliers (Chen et al., 2022b), or ad exposures. The context $\mathbf{X}_i$ captures relevant features such as customer profiles and external confounders (Qiang and Bayati, 2016; Chen et al., 2022a; Simchi-Levi and Wang, 2025). While $\mathcal{D}_n$ can originate from randomized (Manski, 2004) or adaptive experiments (Luckett et al., 2020), high-stakes domains often prohibit experimentation. Consequently, this paper focuses on *off-policy learning* from observational records (Athey and Wager, 2021).

Modern off-policy learning often suffers from data scarcity. The inclusion of non-traditional features creates high-dimensional contexts, making it challenging to credibly estimate counterfactuals (Kallus and Zhou, 2021) required for policy learning. This means a single firm’s in-house dataset is rarely sufficient. To address these constraints, firms increasingly share data, even among direct competitors. Such collaborative pooling constructs comprehensive databases across diverse industries. For example, because commercial insurance risks are highly heterogeneous, major U.S. insurers regularly pool their records at the Insurance Services Office (ISO) to build representative datasets (Frees et al., 2011). Similar collaborative pooling occurs in robotics training (e.g., Open X-Embodiment; O’Neill et al., 2024) and advertising technology via secure data clean rooms such as Snowflake and AWS (Li et al., 2024; Gupta, 2025).

However, learning policies from pooled data introduces two major challenges. First, it induces a distribution shift, as a firm’s target deployment distribution Q often differs substantially from the heterogeneous pooled training distribution P . Second, to protect sensitive customer information, industry standards mandate the use of *differential privacy* (Dwork et al., 2006; Apple Differential Privacy Team, 2017). Because this framework injects calibrated noise before sharing, the system only observes privacy-perturbed contexts, treatments, and outcomes, inherently degrading sample efficiency.

1.1 Our Contribution

In summary, modern policy learning faces a fundamental trilemma: incorporating *high-dimensional contexts* improves performance but necessitates pooling external data, which introduces *distribution shifts* and requires *privacy* mechanisms that reduce effective sample sizes. We investigate this trilemma by formally quantifying these trade-offs using statistical minimax theory.

Assuming the outcome given context and treatment is β -Hölder smooth and invariant across environments, our problem formulation aligns closely with recent literature (Wang et al.,

2025; Ma et al., 2025). However, we tackle the more constrained setting of relying solely on observational data and adapt a significantly less stringent training data coverage condition. Our analysis proceeds from non-private formulations (Section 2) to private extensions (Section 3). In summary, our main contributions are as follows:

- **Relaxed Assumptions for Observational Data Coverage:** Prior studies quantify the difficulty of policy learning using global coverage, evaluating how well observational data spans all possible treatments for a given context. In contrast, we formalize problem difficulty through *local* coverage near the optimal treatment for given context, parameterized as ζ . Crucially, we demonstrate that the fundamental learning trade-off depends solely on this localized coverage.
- **Minimax Lower Bounds for Off-policy Learning:** We characterize the fundamental lower bounds on policy sub-optimal gap, which are unavoidable for any learning algorithm. In the non-private setting and ε -local differential privacy setting (defined in Section 3), sub-optimal gap scales as

$$\Omega\left((\zeta n)^{-\frac{\beta}{2\beta+\alpha+d+1}}\right) \quad \text{and} \quad \Omega\left(\left(n\varepsilon^2\zeta^2\right)^{-\frac{\beta}{2(\alpha+d+\beta+1)}}\right),$$

respectively. These bounds explicitly reveal the inherent trade-offs among sample size n , data dimension d , privacy level ε , and training data coverage relative to the deployment environment (α for covariates, ζ for treatments).

- **Pessimistic Policy Learning Algorithm:** To match these fundamental limits, we propose an algorithm grounded in pessimism, which robustly handles worst-case distributional uncertainty, both under non-private and private setting. This approach avoids the complex task of estimating historical treatment assignment mechanism from a possibly heterogeneous mixture of pooled data. By establishing sub-optimal gap upper bounds that match our lower bounds, we demonstrate that this algorithm achieves minimax optimality in non-private setting (we are currently working on private settings), fully characterizing the fundamental trilemma of off-policy learning under covariate shift.

1.2 Related Work

Off-policy learning is frequently approached by first defining an off-policy evaluation (OPE) objective for a given policy, and then optimizing this estimator across a designated policy class. In the absence of distribution shift or privacy constraints, OPE methods typically fall into three categories: direct methods (DM), inverse propensity weighting (IPW), and doubly robust (DR) estimators. All of these approaches rely on estimating nuisance parameters to correct the bias stemming from non-random treatment assignment.

Specifically, the DM uses regression to model the conditional expected outcome, $\mathbb{E}[Y(t) \mid \mathbf{X}]$. Conversely, IPW estimators reweight the observed outcomes using the inverse of the logging policy’s propensity score (Horvitz and Thompson, 1952; Li et al., 2011). Both foundational methods have notable limitations: the DM is highly sensitive to model misspecification and performs poorly when the logged treatment distribution diverges substantially from the target policy (Beygelzimer and Langford, 2009), while IPW estimators frequently suffer from high variance due to extreme propensity weights.

DR estimators combine the strengths of both approaches by using the DM as a baseline and applying IPW to the regression residuals to correct for remaining bias. Because this hybrid method mitigates inconsistency when the outcome model is misspecified, it is widely adopted for off-policy learning, even under distribution shift (Uehara et al., 2020; Zhao et al., 2025).

Several recent works directly motivate our approach. Pessimistic or conservative methods have proven effective at improving data efficiency in off-policy learning under covariate shift (Park and Lee, 2024). Furthermore, nonparametric regression techniques—particularly local polynomial methods—have been utilized to characterize transferability and adaptation across differing training and test distributions (Kpotufe and Martinet, 2021). Fan et al. (2025) recently integrated these ideas into discrete-action off-policy learning. In this work, we extend their framework to accommodate continuous treatments while simultaneously enforcing local differential privacy constraints.

2 Fundamental Trade-offs in Off-Policy Learning under Covariate Shift

This section investigates the trade-offs between dimensionality and distribution shift in non-private off-policy learning. Section 2.1 formalizes the data-generating process and problem difficulty parameters to establish the minimax lower bound. Section 2.2 then introduces a pessimistic policy learning algorithm with a matching sub-optimal gap upper bound, fully characterizing the minimax rate.

2.1 Problem Formulation and Fundamental sub-optimal gap Lower Bound

Following the off-policy learning literature (Cai et al., 2024; Huang et al., 2023; Kallus and Zhou, 2018; Lange et al., 2012; Levine et al., 2020), We assume that the underlying data (triplets defined in (1)) are drawn identically and independently from P :

$$(\mathbf{X}_i, T_i, Y_i(\cdot)) \stackrel{\text{i.i.d.}}{\sim} P \quad \text{for } i = 1, \dots, n.$$

We recall that the learning algorithm only observes $\mathcal{D}_n = \{(\mathbf{X}_i, T_i, Y_i)\}_{i=1}^n$, where $Y_i = Y_i(T_i)$. To ensure identifiability and regularity of the learning problem, we impose the following structural assumptions on P :

Assumption 1 (Training Data Distribution). For the underlying data distribution P , denote the conditional expected outcome function as $f(\mathbf{x}, t) := \mathbb{E}_P[Y(t) \mid \mathbf{X} = \mathbf{x}]$. We assume:

1. **Unconfoundedness:** For every treatment level $t \in \mathcal{T}$, the potential outcomes are independent of the assigned treatment given the context, that is, $Y(t) \perp\!\!\!\perp T \mid \mathbf{X}$.
2. **Sub-Gaussian Outcomes:** For every $t \in \mathcal{T}$, the potential outcome $Y(t)$ is 1-sub-Gaussian conditional on $\mathbf{X}$.
3. **Hölder Continuity:** The function f is β -Hölder continuous for some $\beta \in (0, 1]$. That is, there exists a constant $C_H > 0$ such that for all $(\mathbf{x}_1, t_1), (\mathbf{x}_2, t_2) \in \mathcal{X} \times \mathcal{T}$,

$$|f(\mathbf{x}_1, t_1) - f(\mathbf{x}_2, t_2)| \leq C_H \|(\mathbf{x}_1, t_1) - (\mathbf{x}_2, t_2)\|_\infty^\beta.$$

We do not impose structural assumptions on the form of historical treatment assignment distribution $P_{T|\mathbf{X}}$, allowing for a highly general policy class. For instance, if $\mathcal{D}_n$ is a pooled

dataset, $P_{T|\mathbf{X}}$ could be a complex hierarchical policy. Because our learning algorithm proposed in Section 2.2 does not require estimating $P_{T|\mathbf{X}}$, it readily accommodates such complexity. The Hölder smoothness assumption is milder than those used in prior work on private dynamic pricing (Song and Jian, 2025; Lei et al., 2024; Chen et al., 2022b) and causal dynamic pricing (Simchi-Levi and Wang, 2025), and is consistent with common assumptions in the study of continuous treatments (Nie et al., 2020). Under assumption 1, we have

$$f(\mathbf{x}, t) = \mathbb{E}[Y(t) \mid \mathbf{X} = \mathbf{x}] = \mathbb{E}[Y_i \mid X_i = \mathbf{x}, T_i = t].$$

Upon deployment in the production environment, the learned policy $\hat{\pi}$ encounters new context $\mathbf{X}^Q$ drawn from a target distribution $Q_{\mathbf{X}}$ supported on $\mathcal{X}$ and selects a treatment $\hat{\pi}(\mathbf{X}^Q) \in \mathcal{T}$, resulting in the outcome $Y(\hat{\pi}(\mathbf{X}^Q))$. We assume a standard covariate shift setting (Shimodaira, 2000; Kpotufe and Martinet, 2021): while the marginal distribution of the contexts differs from the training data ($Q_{\mathbf{X}} \neq P_{\mathbf{X}}$), the conditional expected outcome mechanism remains invariant across both environments: $P_{Y(t)|\mathbf{X}} = Q_{Y(t)|\mathbf{X}}$. Such shifts arise naturally in practical applications. Often, the underlying mechanism mapping inputs to outcomes (such as fundamental human behavior) remains invariant, even as the distribution of the inputs themselves varies across different locations, products, or promotional offers.

The goal of policy learning in this setting is to identify a policy that maximizes the expected outcome, or *policy value*, when applied to the target population. Formally, the value of a policy π is defined as

$$V(\pi) := \mathbb{E}[Y(\pi(\mathbf{X}^Q))] = \mathbb{E}[\mathbb{E}_{Q_{Y(t)|\mathbf{X}}}[Y(\pi(\mathbf{X}^Q)) \mid \mathbf{X}^Q]] = \mathbb{E}_{P, Q_{\mathbf{X}}}[f(\mathbf{X}^Q, \pi(\mathbf{X}^Q))].$$

We define an optimal policy π^* as any policy that maximizes the expected value, meaning $\pi^* \in \arg \max_{\pi} V(\pi)$. Our policy learning goal is equivalently to minimize the policy sub-optimal gap, given by $V(\pi^*) - V(\hat{\pi})$.

Next, we introduce two parameters that quantify the difficulty of the policy learning problem. Both parameters measure the extent to which the training distribution P explores the space required for effective learning. We first formalize the context distributional overlap between $P_{\mathbf{X}}$ and $Q_{\mathbf{X}}$ using the transfer exponent framework (Kpotufe and Martinet, 2021).

Definition 1 (Context Coverage: Transfer Exponent). Let $B_h(\mathbf{x}) \subset \mathcal{X}$ be an L_{∞} -ball of radius h centered at $\mathbf{x}$. We say the transfer exponent from $P_{\mathbf{X}}$ to $Q_{\mathbf{X}}$ is α if there exist constants $\alpha \geq 0$, $\bar{h} > 0$, and $C_{\alpha} > 0$ such that for all $\mathbf{x} \in \text{supp}(Q_{\mathbf{X}})$ and all $h \in (0, \bar{h}]$,

$$P_{\mathbf{X}}(B_h(\mathbf{x})) \geq C_{\alpha} h^{\alpha} Q_{\mathbf{X}}(B_h(\mathbf{x})). \quad (3)$$

Arbitrary distribution pairs trivially satisfy this condition with $\alpha = \infty$. Setting $\alpha = 0$ implies a perfect overlap, guaranteeing dense coverage of $P_{\mathbf{X}}$ wherever the $Q_{\mathbf{X}}$ has mass. Practically, if we think of $B_h(\mathbf{x})$ as a customer segment, Definition 1 quantifies the discrepancy between this segment’s prevalence in historical versus target markets. Rather than assuming a constant baseline of coverage, α allows the coverage bound to relax for niche segments with smaller h . A smaller α implies stronger structural market alignment, which intuitively makes policy learning easier.

Next, we formalize treatment coverage. Effective learning requires sufficient exploration of context-treatment pairs in the training data. To quantify this, we introduce the parameter $\zeta \in [0, 1]$, which captures the propensity of the historical treatment assignment mechanism, $P_{T|\mathbf{X}}$, to explore near-optimal treatments.

Definition 2 (Treatment Coverage: Near-Optimal Treatment Exploration). Let $P_{\mathbf{x},T}$ denote the joint distribution of the context and treatment under P , and let $B_h(\mathbf{x}) \subset \mathcal{X}$ be an L_∞ -ball of radius h centered at $\mathbf{x}$. We say the training distribution P has an exploration coefficient $\zeta \in [0, 1]$ if, for every context $\mathbf{x} \in \text{supp}(P_{\mathbf{x}})$, there exists at least one optimal treatment $t^\dagger(\mathbf{x}) \in \arg \max_{t \in \mathcal{T}} f(\mathbf{x}, t)$ satisfying:

$$\inf_{h \in (0, 1/2]} \frac{P_{\mathbf{x},T}(B_h(\mathbf{x}) \times [t^\dagger(\mathbf{x}) - h, t^\dagger(\mathbf{x}) + h])}{2hP_{\mathbf{x}}(B_h(\mathbf{x}))} \geq \zeta.$$

Returning to the pricing example where $B_h(\mathbf{x})$ represents a customer segment, Definition 2 simply measures the proportion of historical transactions within this segment that offered near-optimal treatments. This localized coverage is strictly weaker than standard assumptions (e.g., Wang et al., 2025), which require strictly positive exploration across the entire treatment space for given context.

Building on the difficulty parameters established in Definitions 1 and 2, we now present the fundamental, unavoidable sub-optimal gap floor that no off-policy learning algorithm can circumvent. The full proof is deferred to Appendix C.

Theorem 1 (Minimax Lower Bound). *If $\zeta n > 1$, then there exists a data-generating instance satisfying Assumption 1 such that for any policy π , the sub-optimal gap satisfies:*

$$V(\pi^*) - V(\pi) = \Omega\left((\zeta n)^{-\frac{\beta}{2\beta + \alpha + d + 1}}\right). \quad (4)$$

2.2 Minimax Optimal Pessimistic Policy Learning Algorithm

Our policy learning algorithm extends the direct method (DM) from Section 1.2 by leveraging regression misspecification to guide policy learning rather than viewing it as a pure error source. We estimate the conditional outcome via nonparametric regression; while its flexibility effectively controls bias, it introduces higher variance, particularly for context-treatment pairs underexplored by P . We formalize this via a pointwise error bound, which allows our policy to select treatments by maximizing a pessimistic lower confidence bound. This inherently penalizes underexplored treatments, mitigating risks associated with data scarcity.

We first define the neighborhood indicator and observation count for a target context $\mathbf{x}^Q \in \mathcal{X}$ and candidate treatment $t \in \mathcal{T}$ at bandwidth $h > 0$:

$$I_{h,i}(\mathbf{x}^Q, t) := \mathbb{1}\{\max(\|\mathbf{X}_i - \mathbf{x}^Q\|_\infty, |T_i - t|) \leq h\}, \quad N_h(\mathbf{x}^Q, t) := \sum_{i=1}^n I_{h,i}(\mathbf{x}^Q, t).$$

Using these definitions, we construct the pessimistic estimator for the conditional outcome, which is based on local constant estimator:

Proposition 1 (Pessimistic Conditional Outcome Estimator). *Let $h > 0$ be a bandwidth parameter and $\beta > 0$ the Hölder smoothness constant with corresponding multiplier C_H . For a target context $\mathbf{x}^Q \in \mathcal{X}$ and treatment $t \in \mathcal{T}$, define the pessimistic estimator as:*

$$\tilde{f}_h(\mathbf{x}^Q, t) := \left(\frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, t)}{N_h(\mathbf{x}^Q, t)} - \sqrt{\frac{2 \log 2n}{N_h(\mathbf{x}^Q, t)}} - C_H h^\beta \right) \mathbb{1}(N_h(\mathbf{x}^Q, t) > 0). \quad (5)$$

Under Assumption 1, we have $\tilde{f}_h(\mathbf{x}^Q, t) \leq f(\mathbf{x}^Q, t)$ with probability at least $1 - 1/n$.

The proof of Proposition 1 is provided in Appendix E.1. Building on this estimator, our policy selects the treatment $\hat{\pi}_h(\mathbf{x}^Q)$ that maximizes the pessimistic lower bound, breaking ties by selecting the smallest treatment value:

$$\hat{\pi}_h(\mathbf{x}^Q) := \min \left\{ \arg \max_{t \in \mathcal{T}} \tilde{f}_h(\mathbf{x}^Q, t) \right\}. \quad (6)$$

Crucially, this policy does not require continuous optimization. Because the indicator function $I_{h,i}(\mathbf{x}^Q, t)$, as a function of t , only changes values when $t = T_i \pm h$, the estimator $\tilde{f}_h(\mathbf{x}^Q, t)$ is piecewise constant in t for any fixed $\mathbf{x}^Q$. Consequently, the exact global maximum of $\tilde{f}_h(\mathbf{x}^Q, t)$ can be computed efficiently via a linear scan over at most $2n + 2$ candidate points: the domain endpoints $\{0, 1\}$ and the empirical jump points $\{\max(0, T_i - h), \min(1, T_i + h)\}_{i=1}^n$.

As established in the following theorem (proof deferred to Appendix E.3), this pessimistic estimator achieves minimax optimality by attaining a sub-optimal gap upper bound that matches the lower bound of Theorem 1 up to a constant.

Theorem 2. *Under Assumption 1, there exists a bandwidth $h > 0$ such that the policy defined in (6) achieves*

$$V(\pi^*) - V(\hat{\pi}_h) = \tilde{O}\left((\zeta n)^{-\frac{\beta}{2\beta + \alpha + d + 1}}\right), \quad (7)$$

for sufficiently large n , where $\tilde{O}(\cdot)$ hides logarithmic factors.

3 Extension to Local Differential Privacy

Building upon the covariate shift setting studied in Section 2, we now assume the training data is subject to *local differential privacy* (LDP), introducing an additional layer of complexity. Section 3.1 formally introduces the privacy framework and learning objective, establishing the minimax lower bound for this setting. Subsequently, Section 3.2 presents a private pessimistic learning algorithm, extending the core learning strategy developed in Section 2.2.

3.1 Problem Formulation and Fundamental sub-optimal gap Lower Bound

Local differential privacy (LDP) is a widely adopted framework for ensuring user-level data protection for decision-making at scale, famously deployed by companies such as Google and Apple (Mun et al., 2025). To safeguard individual user’s data, the logged contexts, treatments, and outcomes are randomized via a privacy mechanism governed by a budget ε . Modeled as a Markov kernel $M_i(\cdot \mid \mathbf{x}, t, y)$, this mechanism ensures strong per-sample protection (Definition 3) but distorts the underlying distributions, restricting the learner to the privatized dataset:

$$\tilde{\mathcal{D}}_n := \left\{ M_i(\mathbf{X}_i, T_i, Y_i) \right\}_{i=1}^n. \quad (8)$$

Definition 3 (Local Differential Privacy). Given a privacy budget $\varepsilon > 0$, let the true data tuple $(\mathbf{X}_i, T_i, Y_i)$ take values in the measurable space $(\mathcal{X} \times \mathcal{T} \times \mathcal{Y}, \mathcal{G})$, and let its privatized view $M_i(\mathbf{X}_i, T_i, Y_i)$ take values in $(\mathcal{M}_i, \mathcal{B}_i)$. We say $M_i(\mathbf{X}_i, T_i, Y_i)$ satisfies ε -local differential privacy (ε -LDP) if the privacy mechanism M_i is a Markov kernel from $(\mathcal{X} \times \mathcal{T} \times \mathcal{Y}, \mathcal{G})$ to $(\mathcal{M}_i, \mathcal{B}_i)$ such that for all measurable sets $A \in \mathcal{B}_i$ and all possible data pairs $(\mathbf{x}, t, y)$ and $(\mathbf{x}', t', y')$ in $\mathcal{X} \times \mathcal{T} \times \mathcal{Y}$, the following inequality holds:

$$M_i(A \mid \mathbf{x}, t, y) \leq e^\varepsilon M_i(A \mid \mathbf{x}', t', y').$$

To quantify the difficulty of the private learning problem, we adapt the parameters from the non-private setting to accommodate the binning mechanism that we will use for local differential privacy. Specifically, we redefine the transfer exponent (Definition 1) and exploration coefficient (Definition 2) to operate over fixed partitions of the context and treatment spaces, rather than over arbitrary neighborhoods centered at individual data points.

Definition 4 (Partition-Based Transfer Exponent). Let $\mathcal{A}_h$ denote a partition of $\mathcal{X}$ into bins of width h . We say the partition-based transfer exponent from $P_{\mathbf{X}}$ to $Q_{\mathbf{X}}$ is α if there exist constants $\alpha \geq 0$ and $C_\alpha > 0$ such that for any bin $A \in \mathcal{A}_h$, we have:

$$P_{\mathbf{X}}(A) \geq C_\alpha h^\alpha Q_{\mathbf{X}}(A).$$

Definition 5 (Partition-Based Exploration Coefficient). Let $P_{\mathbf{X},T}$ denote the joint distribution of the context and treatment under P . For any target context $\mathbf{x}^Q \in \mathcal{X}$, let $A_{h,\mathcal{X}}(\mathbf{x}^Q)$ denote the context bin of width h containing $\mathbf{x}^Q$, and let $A_{h,\mathcal{T}}(t)$ denote the treatment bin containing a given treatment t . We define the exploration coefficient $\zeta \in (0, 1]$ as a constant such that for every $\mathbf{x}^Q \in \text{supp}(P_{\mathbf{X}})$, there exists at least one optimal treatment $t^\dagger(\mathbf{x}^Q) \in \arg \max_{t \in \mathcal{T}} f(\mathbf{x}^Q, t)$ satisfying:

$$\inf_{h \in (0, 1/2]} \frac{P_{\mathbf{X},T}(A_{h,\mathcal{X}}(\mathbf{x}^Q) \times A_{h,\mathcal{T}}(t^\dagger(\mathbf{x}^Q)))}{2h P_{\mathbf{X}}(A_{h,\mathcal{X}}(\mathbf{x}^Q))} \geq \zeta.$$

Theorem 3 establishes the minimax lower bound on sub-optimal gap for any ε -LDP policy learning algorithm. The full proof is deferred to Appendix D. While it mirrors the Fano-type argument used in the non-private setting, it fundamentally relies on the contraction of KL divergence between distributions induced by the privacy mechanism.

Theorem 3 (LDP Minimax Lower Bound). *Assume $\varepsilon \in (0, 1]$ and $n\varepsilon^2\zeta^2 > 1$. Under Assumption 1, for any ε -LDP mechanism $\mathcal{M}$ and any policy $\hat{\pi}$ trained from its output, there exists a data-generating instance such that the sub-optimal gap satisfies:*

$$V(\pi^*) - V(\hat{\pi}) = \Omega \left(\left(n\varepsilon^2\zeta^2 \right)^{-\frac{\beta}{2(\alpha+d+\beta+1)}} \right). \quad (9)$$

3.2 Private Pessimistic Policy Learning Algorithm

This section presents a privacy mechanism and a pessimistic learning algorithm that operates on the privatized data, extending the core strategy developed in Section 2.2.

To implement the privacy mechanism, the joint space $\mathcal{X} \times \mathcal{T}$ is partitioned into $K := \lfloor h^{-(d+1)} \rfloor$ equal-sized hypercubes of side length h , denoted by $\{A_{h,j}\}_{j=1}^K$. Each user i then privatizes their data tuple $(\mathbf{X}_i, T_i, Y_i)$ in two steps. First, the user maps the context-treatment pair to a K -dimensional one-hot vector $\mathbf{U}_i \in \{0, 1\}^K$, where the j -th component is defined as:

$$U_{i,j} = \mathbb{1}\{(\mathbf{X}_i, T_i) \in A_{h,j}\}$$

Next, the user injects independent Laplace noise, releasing the privatized vectors:

$$\tilde{\mathbf{U}}_i = \mathbf{U}_i + \boldsymbol{\zeta}_i \quad \text{and} \quad \tilde{\mathbf{V}}_i = Y_i \mathbf{U}_i + \boldsymbol{\xi}_i, \quad (10)$$

where the noise components $\zeta_{i,j}$ and $\xi_{i,j}$ are drawn independently from a zero-mean Laplace distribution with variance $32/\varepsilon^2$. As formally established by Berrett et al. (2021, Proposition 1), this additive noise mechanism guarantees ε -local differential privacy.

To learn a policy from the privatized dataset $\{\tilde{\mathbf{U}}_i, \tilde{\mathbf{V}}_i\}_{i=1}^n$, we once again construct a pessimistic estimator for the conditional outcome. We build this in three stages, defining an intermediate non-private estimator, its conditional expectation, and the final private estimator.

Let $j(\mathbf{x}^Q, t^Q)$ denote the index of the specific bin containing a query point $(\mathbf{x}^Q, t^Q) \in \mathcal{X} \times \mathcal{T}$. For each user i , the privacy mechanism internally generates a true indicator vector $\mathbf{U}_i \in \mathbb{R}^K$ that is strictly unobservable to the learning algorithm. Using this, Definition 6 defines a local constant estimator similar to the non-private case. The distinction is that this private mechanism restricts estimation to the fixed partition $\{A_{h,j}\}_{j=1}^K$, structurally preventing the algorithm from centering neighborhoods exactly at the query point.

Definition 6 (Non-Private Estimator). Given a query point $(\mathbf{x}^Q, t^Q) \in \mathcal{X} \times \mathcal{T}$, the non-private estimator $\hat{f}_n(\mathbf{x}^Q, t^Q)$ computes the average outcome of all samples falling into the same bin as the query point:

$$\hat{f}_n(\mathbf{x}^Q, t^Q) := \frac{\sum_{i=1}^n Y_i U_{i,j(\mathbf{x}^Q, t^Q)}}{\sum_{i=1}^n U_{i,j(\mathbf{x}^Q, t^Q)}}. \quad (11)$$

If no observations fall into the bin, we adopt the standard convention $0/0 = 0$.

Because the $\mathbf{U}_i$ and Y_i are unobservable to the learning algorithm, this non-private estimator cannot be computed directly. Instead, Definition 7 constructs a private counterpart by replacing the numerator and denominator with their noise-injected equivalents.

Definition 7 (Private Estimator). Given a query point $(\mathbf{x}^Q, t^Q) \in \mathcal{X} \times \mathcal{T}$, the private estimator $\tilde{f}_{n,\varepsilon}(\mathbf{x}^Q, t^Q)$ is computed exclusively using the privatized ε -LDP vectors:

$$\tilde{f}_{n,\varepsilon}(\mathbf{x}^Q, t^Q) := \frac{\sum_{i=1}^n \tilde{V}_{i,j(\mathbf{x}^Q, t^Q)}}{\sum_{i=1}^n \tilde{U}_{i,j(\mathbf{x}^Q, t^Q)}}. \quad (12)$$

Because the Laplace noise is continuous, the denominator is non-zero almost surely.

Theorem 4 demonstrates that while the private estimator is inherently noisy, it is correctly centered around the expected non-private estimator. Crucially, the magnitude of this deviation can be bounded using only the observed privatized information (proof deferred to Appendix F.1).

Theorem 4 (Concentration of the Private Estimator). *Let us denote the expected non-private estimator as*

$$\bar{f}_n(\mathbf{x}^Q, t^Q) := \mathbb{E}_{P_{Y(t)|\mathbf{x}}} [\hat{f}_n(\mathbf{x}^Q, t^Q)] = \frac{\sum_{i=1}^n f(\mathbf{X}_i, T_i) U_{i,j(\mathbf{x}^Q, t^Q)}}{\sum_{i=1}^n U_{i,j(\mathbf{x}^Q, t^Q)}}. \quad (13)$$

Suppose $\varepsilon \leq 8\sqrt{2n/\log n}$. For a given query point $(\mathbf{x}^Q, t^Q)$, let $\tilde{N}_h(\mathbf{x}^Q, t^Q) := \sum_{i=1}^n \tilde{U}_{i,j(\mathbf{x}^Q, t^Q)}$ represent the observed noisy observation count within the respective bin. With probability at least $1 - 6/n$, the error of the private estimator is bounded by:

$$|\tilde{f}_{n,\varepsilon}(\mathbf{x}^Q, t^Q) - \bar{f}_n(\mathbf{x}^Q, t^Q)| = O\left(\frac{\sqrt{\log n \max\{\tilde{N}_h(\mathbf{x}^Q, t^Q), n/\varepsilon^2\}}}{|\tilde{N}_h(\mathbf{x}^Q, t^Q)|}\right). \quad (14)$$

Leveraging the concentration bound from Theorem 4 and the geometric bias derived from the Hölder smoothness assumption, we construct a pessimistic estimator. The full proof is deferred to Appendix F.2.

Lemma 1 (LDP Pessimistic Estimator). *Define the stability threshold $\tau_{n,\varepsilon} \asymp \sqrt{n \log n / \varepsilon^2}$, and the privacy-adjusted confidence width:*

$$W_{n,\varepsilon,h}(\mathbf{x}^Q, t^Q) := C_{\text{LDP}} \frac{\sqrt{\max\{n/\varepsilon^2, \tilde{N}_h(\mathbf{x}^Q, t^Q)\} \log n}}{\tilde{N}_h(\mathbf{x}^Q, t^Q)}.$$

The **pessimistic LDP estimator** penalizes the private estimator by this confidence width and the geometric bias:

$$\underline{f}_{n,\varepsilon,h}(\mathbf{x}^Q, t^Q) := \mathbb{1}\{\tilde{N}_h(\mathbf{x}^Q, t^Q) \geq \tau_{n,\varepsilon}\} \left(\tilde{f}_{n,\varepsilon}(\mathbf{x}^Q, t^Q) - W_{n,\varepsilon,h}(\mathbf{x}^Q, t^Q) - C_H h^\beta \right). \quad (15)$$

With probability at least $1 - 6/n$, this estimator satisfies the pessimism property: $\underline{f}_{n,\varepsilon,h}(\mathbf{x}^Q, t^Q) \leq f(\mathbf{x}^Q, t^Q)$.

Building on this pessimistic estimator, the policy $\hat{\pi}_{h,\varepsilon}$ maps a given context $\mathbf{x}^Q$ to a treatment by maximizing the privacy-adjusted lower confidence bound, breaking ties arbitrarily:

$$\hat{\pi}_{h,\varepsilon}(\mathbf{x}^Q) \in \operatorname{argmax}_{t \in \mathcal{T}} \underline{f}_{n,\varepsilon,h}(\mathbf{x}^Q, t).$$

Because the estimator $\underline{f}_{n,\varepsilon,h}$ is piecewise constant over each bin $A_{h,j}$, for any given context $\mathbf{x}^Q$, the exact global maximum is easily found via a discrete search; the algorithm simply isolates the bins whose context domains contain $\mathbf{x}^Q$ and scans across that finite subset along the treatment axis.

We conjecture that this private pessimistic policy learning algorithm achieves minimax optimality, as formalized below.

Conjecture 1 (Suboptimality of the LDP Policy). Provided Assumption 1 holds and the bandwidth h is chosen optimally, the suboptimality gap of the LDP pricing policy $\hat{\pi}_{h,\varepsilon}$ satisfies:

$$V(\pi^*) - V(\hat{\pi}_{h,\varepsilon}) = \tilde{O} \left(\left(n \varepsilon^2 \zeta^2 \right)^{-\frac{\beta}{2\beta+2(\alpha+d+1)}} \right). \quad (16)$$

4 Conclusion

In this work, we formalized the fundamental trade-offs in off-policy learning under the simultaneous challenges of covariate shift and local differential privacy. By characterizing the difficulty of the learning problem through a localized treatment coverage parameter ζ , rather than relying on restrictive global overlap assumptions, we established rigorous minimax lower bounds for both non-private and ε -LDP settings. To achieve these theoretical limits, we introduced a unified pessimistic policy learning framework that robustly penalizes both underexplored treatments and privacy-induced noise. We proved that this approach yields a minimax-optimal policy in the non-private regime, fully charting the sample complexity required to overcome data dimensionality and distribution shift without the need to explicitly model heterogeneous historical logging policies.

Our future goal is the formal resolution of Conjecture 1, which posits that our privacy-adjusted pessimistic estimator achieves strict minimax optimality under ε -LDP constraints. Successfully proving this will complete the theoretical picture of the privacy-shift-dimensionality trilemma.

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A List of Notations

A.1 training Distribution

- $\mathcal{D} := \{\mathbf{X}_i, T_i, Y_i\}_{i=1}^n$: training data triplets; random vector in $[0, 1]^d \times [0, 1] \times [0, 1]$
- P : training context distribution.
- p : density of the training context distribution P . Used in minimax lower bound proof.
- t : realization of training treatment.
- μ : joint distribution of training context-treatment pair.
- M_i : privacy mechanism applied to i th training observation

A.2 Constants

- $C_{\text{privateTV}}$: Lemma 18

B Technical lemmas and definitions

The following lemma provides a bound on the expectation of the reciprocal of a binomial random variable, restricted to the event that it is positive.

Lemma 2 (Lemma 6 in Pathak et al., 2022). *Let n be a positive integer and $t \in (0, 1)$. Suppose $U \sim \text{Bin}(n, p)$. Then,*

$$\mathbb{E} \left[\frac{1}{U} \cdot \mathbb{1}\{U > 0\} \right] \leq \frac{4}{\mathbb{E}[U]}.$$

Lemma 3 (Lemma 15 in Cai et al. 2024). *For any $a, b \in [0, 1]$, let $\text{Bern}(a)$ and $\text{Bern}(b)$ denote two Bernoulli distributions with parameters a and b , respectively. Then one has*

$$KL(\text{Bern}(a) \parallel \text{Bern}(b)) \leq \frac{(a - b)^2}{b(1 - b)}. \quad (17)$$

In addition, if $|b - 1/2| \leq 1/4$, then one further has

$$KL(\text{Bern}(a) \parallel \text{Bern}(b)) \leq 8(a - b)^2. \quad (18)$$

C Proof of Theorem 1

This section proves the non-private minimax lower bound presented in Theorem 1. We use the following version of Fano's Lemma (proof in Appendix C.5) which does not require a metric.

Theorem 5 (Fano's Lemma). *Let Θ be a class of problem instances (distributions). Consider a loss function $L : \Theta \times \mathcal{T} \rightarrow \mathbb{R}_+$, which evaluates the quality of a treatment for a given problem instance. If there exists a finite number of problem instances $\theta_1, \dots, \theta_m \in \Theta$ such that*

$$L(\theta_j, t) + L(\theta_k, t) \geq \Delta, \quad \forall j \neq k \in [m], \quad t \in \mathcal{T},$$

the minimax risk is bounded from below as the following:

$$\begin{aligned} \inf_{\pi} \sup_{\theta \in \Theta} \mathbb{E}_{D \sim \theta} [L(\theta, \pi(D))] &\geq \frac{\Delta}{2} \inf_{\Psi} \frac{1}{m} \sum_{j=1}^m \theta_j(\Psi \neq j) \\ &\geq \frac{\Delta}{2} \left\{ 1 - \frac{1}{\log m} \left(\frac{1}{m^2} \sum_{j,k=1}^m KL(\theta_j \parallel \theta_k) + \log 2 \right) \right\} \end{aligned}$$

where the infimum on the first term is taken over all pricing policies π , while the infimum in the second term is taken over all measurable tests $\Psi : \mathcal{X} \rightarrow [m]$.

The proof follows the following intermediate steps:

1. Construction of problem instances (Appendix C.1),
2. Computation in distribution space: upper bounding the KL divergence (Appendix C.2)
3. Computation in parameter space: lower bounding the sub-optimal gap (Appendix C.3)
4. Application of Fano's lemma (Appendix C.4).

C.1 Construction of Problem Instances

The construction of the problem instances relies centrally on a radius parameter $r \in (0, 1/2)$. This parameter dictates three key components of the proof:

1. The discretization of the domains $\mathcal{X}$ and $\mathcal{T}$ into grids with spacing determined by r (Appendix C.1.1).
2. The construction of an index set for the packing argument, yielding a finite, well-separated collection of distributions that satisfies the conditions of Theorem 5 (Definition 20).
3. The formulation of the underlying distributions (Appendix C.1.2) and the corresponding outcome functions (Appendix C.1.3) based on these grids and packing indices.

The specific value of the radius r is left unspecified until the conclusion of the proof. In the final step (Appendix C.4), we instantiate r with the precise value that yields the desired rate.

C.1.1 Construction of Domain Grids and Packing Indices

Based on the radius parameter r , this section constructs the context grid (Definition 8), treatment grid (Definition 9), and packing indices (Definition 10).

Definition 8 (Context grid). We partition the context space $\mathcal{X} = [0, 1]^d$ into hypercubes, each with a side length of $2r$. Let $\mathcal{S}_{\mathcal{X},r}$ denote the set of centers for these hypercubes. The coordinates of these grid points start from r and increment with a uniform spacing of $2r$. The cardinality is $\lfloor 1/(2r) \rfloor^d$. More formally:

$$\begin{aligned} \mathcal{S}_{\mathcal{X},r} &= \left\{ \mathbf{x}_1^*, \dots, \mathbf{x}_{\lfloor 1/(2r) \rfloor^d}^* \right\} \\ &:= \left\{ \mathbf{x} = (x_1, x_2, \dots, x_d) \in [0, 1]^d \mid x_i = \left(k_i - \frac{1}{2}\right) 2r, k_i \in \left[\left\lfloor \frac{1}{2r} \right\rfloor\right], \forall i \in [d] \right\}. \end{aligned} \quad (19)$$

Definition 9 (Treatment grid). We partition the treatment space $\mathcal{T} = [0, 1]$ into intervals, each with a length of r . Let $\mathcal{S}_{\mathcal{T},r}$ represent the centers of these intervals. These grid points begin at $r/2$ and increase by increments of r . The cardinality is $\lfloor 1/r \rfloor$. More formally:

$$\mathcal{S}_{\mathcal{T},r} = \left\{ (k - \tfrac{1}{2})r : k \in \left[\left\lfloor \frac{1}{r} \right\rfloor\right] \right\} = \left\{ t_1^*, \dots, t_{\lfloor 1/r \rfloor}^* \right\}. \quad (20)$$

We now construct an index set that will parametrize the hard instances used in our Fano-type argument.

Definition 10 (Packing indices). First, define the integer

$$m := \lfloor c_m/r \rfloor^d \quad (21)$$

for a constant $0 < c_m < 1/2$. Using the Varshamov-Gilbert bound (see, e.g., Example 5.3 of Wainwright 2019), we can find a set of well-separated vectors

$$\Omega_m := \{\boldsymbol{\omega}_j\}_{j=0}^M \subset \{\pm 1\}^m \quad (22)$$

with sufficiently large cardinality and separation:

$$\log_2(M) \geq \frac{m}{8} \quad \text{and} \quad \rho(\boldsymbol{\omega}_j, \boldsymbol{\omega}_k) \geq \frac{m}{8}, \quad \forall 0 \leq j < k \leq M, \quad (23)$$

where $\rho(\boldsymbol{\omega}, \boldsymbol{\omega}')$ is the Hamming distance, which measures the number of indices at which the vectors $\boldsymbol{\omega}$ and $\boldsymbol{\omega}'$ differ.

Based on this index, following a standard technique in nonparametric statistics (see, for example, Example 15.15 in Wainwright (2019) for a detailed exposition), Appendix C.1.3 will construct a local packing of the function space.

C.1.2 Construction of Problem Instances: Shared Components

Using the grids and packing index set constructed in Appendix C.1.1, we define a collection of training and target distributions, denoted by $t_{\boldsymbol{\omega}}$ and $Q_{\boldsymbol{\omega}}$, which are indexed by $\boldsymbol{\omega} \in \Omega_m$. Applying Fano's method requires computing loss function (sub-optimal gap in our case) and distribution distances (KL divergence or TV) between problem instances. Our problem instance constructions are designed make these computations easy.

To distinguish these engineered problem instances, tailored specifically for the Fano-type argument, from the general problem distributions evaluated in the main text, we formalize this collection with respect to their underlying density functions:

$$\mathcal{H}_{\Omega_m} := \{(q, p, \wp, f_{\boldsymbol{\omega}}) \mid \boldsymbol{\omega} \in \Omega_m\}, \quad (24)$$

where q is the target context density, p is the training context density, $\wp$ is the offline logging propensity, and $f_{\boldsymbol{\omega}}$ is the outcome density for both the training and target.

These instances are constructed based on the following design principles:

- Only the outcome density $f_{\boldsymbol{\omega}}$ depends on the packing index $\boldsymbol{\omega} \in \Omega_m$.
- All the other components (q , p , and $\wp$) remain fixed across all instances.

This section constructs the shared parts (q , p , and $\wp$). Appendix C.1.3 defines the index-specific part ($f_{\boldsymbol{\omega}}$).

Definition 11 (Target context density). We define the target context distribution Q by specifying its density, denoted $q(\mathbf{x})$, as $q(\mathbf{x}) = 1$.

We define the training context-price pair distribution μ by first specifying the training context distribution P and then the conditional training price distribution π_{off} .

Definition 12 (Training context density). Pick $C_{\alpha} < 1/3$. The training context distribution part of $t_{\boldsymbol{\omega}, \mathbf{x}}$ is defined by its piecewise constant density, $p(\mathbf{x})$:

$$p(\mathbf{x}) := \begin{cases} C_{\alpha} r^{\alpha}, & \text{if } \mathbf{x} \in \bigcup_{j=1}^m B_{r/2}(\mathbf{x}_j^*), \\ \delta, & \text{if } \mathbf{x} \in \bigcup_{j=1}^m (B_r(\mathbf{x}_j^*) \setminus B_{r/2}(\mathbf{x}_j^*)), \\ 1, & \text{if } \mathbf{x} \in \mathcal{X} \setminus \bigcup_{j=1}^m B_r(\mathbf{x}_j^*). \end{cases}$$

where the constant δ is chosen to ensure that $p(\mathbf{x})$ integrates to one:

$$\delta := \frac{2^d - C_\alpha r^\alpha}{2^d - 1}$$

and α and C_α are the transfer exponent and its corresponding constant, as defined in Definition 1.

The following lemma (proof in Appendix C.6.1) ensures that this $p(\mathbf{x})$ integrates to 1.

Lemma 4. *Regardless of the value of C_α , the density $p(\mathbf{x})$ defined in Definition 12 satisfies $\int_{\mathcal{X}} p(\mathbf{x}) d\mathbf{x} = 1$.*

The following lemma (proof in in Appendix C.6.2) demonstrate that our constructions satisfy the assumptions for the transfer exponent (Definition 1).

Lemma 5 (Transfer exponent). *For the training context distribution $t_{\omega, \mathbf{x}}$ constructed as in Definition 12 and the target context distribution $Q_{\omega, \mathbf{x}}$ constructed as in (11), the transfer exponent of $t_{\omega, \mathbf{x}}$ with respect to $Q_{\omega, \mathbf{x}}$ is α , with corresponding constant C_α .*

Definition 13 (Offline logging policy propensity). The offline logging policy $\pi_{\text{off}}(\cdot | \mathbf{x}_i)$ is a binary policy that is independent of the context, that is, a completely random policy. It is characterized by the generalized propensity (conditional density) $\wp(\cdot | \mathbf{x}_i)$, formally defined as:

$$\wp(t_i | \mathbf{x}_i) := \begin{cases} \zeta, & \text{if } t_i \in [\tilde{t} - r/2, \tilde{t} + r/2], \\ \frac{1 - r\zeta}{1 - r}, & \text{otherwise,} \end{cases}$$

where ζ is the exploration coefficient, and $\tilde{t} \in \mathcal{S}_{\mathcal{T}, r}$ is a constant to be defined later.

and the exploration coefficient (Definition 2).

Lemma 6 (Near-optimal price exploration). *The data collecting policy π_{off} with conditional density $\wp(p | \mathbf{x})$ defined as (13) satisfies the near-optimal price exploration assumption with coefficient ζ .*

Proof of Lemma 6 is provided in Appendix C.6.4.

C.1.3 Construction of the Outcome Distributions

For both target and training data, let the random outcome, conditional on the context $\mathbf{x}$ and treatment t , be a Bernoulli random variable with success probability $f_\omega(\mathbf{x}, t)$. This setting satisfies the 1-sub-Gaussian assumption. The outcome functions $\{f_\omega\}_{\omega \in \Omega_m}$ are constructed as follows. We begin by defining a simple Hölder continuous function.

Lemma 7. *The function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$ defined by*

$$\phi(z) := \begin{cases} 1, & \text{if } 0 \leq z < \frac{1}{4}, \\ (2 - 4z)^\beta, & \text{if } \frac{1}{4} \leq z < \frac{1}{2}, \\ 0, & \text{otherwise,} \end{cases} \quad (25)$$

is Hölder continuous with exponent $\beta \in (0, 1]$ and Hölder constant 4^β .

Proof of Lemma 7 is provided in Appendix C.6.5.

Next, define the function $\varphi_\beta : [0, 1]^d \times \mathcal{T} \rightarrow [0, 1/4]$ via

$$\varphi_\beta(\mathbf{x}, t) := C_\beta r^\beta \phi\left(\frac{1}{r} \max\{\|\mathbf{x}\|_\infty, |t|\}\right), \quad (26)$$

where $C_\beta := \min\{C_H, 1\}/4^\beta$.

We add this φ_β function as bumps, centered at the grid points $S_{\mathcal{X}, r}$ in the context space, and also at a specific point t_u^* within $S_{\mathcal{T}, r}$ in the treatment space. This point t_u^* is defined as follows. Fix a policy π . For any grid point $t_i^* \in S_{\mathcal{T}, r}$, define a indicator variable

$$I_{r, \pi}^Q(t_i^*) := \mathbb{1}\left\{\pi(\mathbf{X}^Q) \in [t_i^* - r/2, t_i^* + r/2]\right\}, \quad (27)$$

which is a transformation of $\mathbf{X}^Q$. For any policy π , we have

$$\sum_{i=1}^{\lfloor 1/r \rfloor} \mathbb{E}_Q[I_{r, \pi}^Q(t_i^*)] = \mathbb{E}_Q\left[\sum_{i=1}^{\lfloor 1/r \rfloor} I_{r, \pi}^Q(t_i^*)\right] = 1.$$

The reason is, the candidate points $t_1^*, t_2^*, \dots, t_{\lfloor 1/r \rfloor}^*$ form a grid with spacing r , starting from $r/2$ (see Definition 9). Therefore, the associated intervals $[t_i^* - r/2, t_i^* + r/2]$ are mutually exclusive (disjoint) and collectively exhaust (cover) the entire valid treatment space $[0, 1]$. Because these intervals form a perfect partition of the space, for any random realization of the context $\mathbf{X}^Q$, the policy output $\pi(\mathbf{X}^Q)$ must fall into exactly one of these intervals, regardless of the distribution $Q_{\mathbf{X}}$.

Therefore, by the pidgeonhole principle, there must exist at least one grid point $t_u^* \in S_{\mathcal{T}, r}$ such that

$$\mathbb{E}_Q[I_{r, \pi}^Q(t_u^*)] \leq \frac{1}{\lfloor 1/r \rfloor}. \quad (28)$$

We denote that t_u^* as $\tilde{t}$. With this definition, we finally construct a Hölder continuous outcome function:

Definition 14 (Outcome distribution). Fix $\beta \in (0, 1]$. For any $\omega \in \Omega_m$, the outcome function $f_\omega : \mathcal{X} \times \mathcal{T} \rightarrow \mathcal{Y}$ is defined as

$$f_\omega(\mathbf{x}, t) := \frac{1}{2} + \sum_{j=1}^m \omega_j \varphi_\beta\left((\mathbf{x} - \mathbf{x}_j^*), (t - \tilde{t})\right) \cdot \mathbb{1}\left\{\mathbf{x} \in B_r(\mathbf{x}_j^*)\right\}. \quad (29)$$

The training and target outcomes, Y and Y^Q , are drawn from Bernoulli distributions with success probabilities $f_\omega(\mathbf{x}_i, t_i)$ and $f_\omega(\mathbf{x}^Q, \pi(\mathbf{x}^Q))$, respectively.

Lemma 8. *The outcome function $f_\omega(\mathbf{x}, t)$ defined in Definition 14 is Hölder continuous with exponent β and Hölder constant $C_\beta > 0$.*

The proof of Lemma 8 is provided in Appendix C.6.6.

C.1.4 Joint Distributions of the Problem Instances

For each problem instance indexed by $\omega \in \Omega_m$, defined in Appendix C.1, its full joint distribution is denoted by $\theta_{\omega, \pi}$, which we formally define as follows.

Definition 15 (Data generating process). Given a policy π , the full problem instance $\theta_{\omega, \pi}$ is a joint distribution of t_ω , Q_ω and π . The data generating process under $\theta_{\omega, \pi}$, and associated notations, are defined as follows.

The training data observation $(\mathbf{x}_i, t_i, y_i)$ is generated i.i.d. from a joint distribution t_ω whose density is factored as:

$$p(\mathbf{x}_i) \wp(t_i | \mathbf{x}_i) \vartheta_{f_\omega}(y_i; \mathbf{x}_i, t_i). \quad (30)$$

Specifically, each observation is drawn via the following sequential process:

1. **Context:** A training context $\mathbf{x}_i$ is drawn from the training context distribution which has density $p(\cdot)$.
2. **Treatment:** A training treatment t_i is drawn conditionally on the context $\mathbf{x}_i$ according to the offline logging policy propensity $\wp(\cdot | \mathbf{x}_i)$.
3. **Outcome:** The outcome y_i is drawn from a Bernoulli distribution governed by the success probability $f_\omega(\mathbf{x}_i, t_i)$. Its conditional likelihood is denoted by $\vartheta_{f_\omega}(\cdot; \mathbf{x}_i, t_i)$.

The target data observation $(\mathbf{x}^Q, \pi(\mathbf{x}^Q), y^Q)$ is generated from the joint distribution $Q_{\omega, \pi}$ whose density is factored as:

$$q(\mathbf{x}^Q) \vartheta_{f_\omega}(y^Q; \mathbf{x}^Q, \pi(\mathbf{x}^Q)). \quad (31)$$

Specifically, each target observation is generated via the following sequential process:

1. **Context:** A target context $\mathbf{x}^Q$ is drawn from the target context distribution Q , which has density $q(\cdot)$.
2. **Treatment:** The target treatment is deterministically set according to the policy as $\pi(\mathbf{x}^Q)$.
3. **Outcome:** The outcome y^Q is drawn from a Bernoulli distribution governed by the success probability $f_\omega(\mathbf{x}^Q, \pi(\mathbf{x}^Q))$. Its conditional likelihood is denoted by $\vartheta_{f_\omega}(\cdot; \mathbf{x}^Q, \pi(\mathbf{x}^Q))$.

C.2 Computation in Distribution Space: Upper Bounding the KL Divergence

Appendix C.2.1 establishes a key structural property of our construction that simplifies the KL divergence computation. Building on this, Appendix C.2.2 leverages this property to compute the KL divergence between problem instances, concluding that

$$\text{KL}(\theta_{\omega', \pi}, \theta_{\omega, \pi}) \lesssim m \zeta n r^{d+1+2\beta+\alpha} + r^{1+2\beta}.$$

C.2.1 Simplifying the KL Divergence via Localized Support

The fundamental mechanism that makes the KL divergence computation tractable is the spatial localization of the outcome functions. Specifically, the perturbed outcome functions $\{f_\omega\}_{\omega \in \Omega_m}$ are identical to the baseline value of $1/2$ over the vast majority of the domain. Lemma 9 formalizes this property.

Lemma 9 (Localized Bumps / Flat Regions). *For any packing index $\omega \in \Omega_m$, the outcome function $f_\omega(\mathbf{x}, t)$ defined in (29) evaluates exactly to $1/2$ whenever the context-treatment pair falls outside the designated perturbation zones. Specifically:*

- (i) **Context isolation:** For any $\mathbf{x} \in \mathcal{X} \setminus \bigcup_{j=1}^m B_{r/2}(\mathbf{x}_j^*)$ and any treatment $t \in [0, 1]$, we have $f_\omega(\mathbf{x}, t) = \frac{1}{2}$.

(ii) **Treatment isolation:** For any treatment $t \in \mathcal{T} \setminus [\tilde{t} - r/2, \tilde{t} + r/2]$ and any context $\mathbf{x} \in \mathcal{X}$, we have $f_\omega(\mathbf{x}, t) = \frac{1}{2}$.

Proof. The proof follows directly from the algebraic construction of the outcome function and is entirely independent of the density components p, q , and $\wp$. Specifically, the base bump function $\phi(z)$ defined in (25) strictly vanishes for $z \geq 1/2$. Because the spatial perturbation $\varphi_\beta(\mathbf{x}, t)$ from (26) is constructed using the ℓ_∞ -norm, evaluating $\max\{\|\mathbf{x} - \mathbf{x}_j^*\|_\infty, |t - \tilde{t}|\}/r$, it evaluates identically to zero whenever the context $\mathbf{x}$ is further than $r/2$ from all grid centers, or when the treatment t is further than $r/2$ from $\tilde{t}$. $\square$

Lemma 9 significantly simplifies the KL divergence computation. Because the outcome function is the sole component that varies across problem instances, the log-likelihood ratio between any two instances vanishes everywhere except within highly localized regions around the active grid points.

C.2.2 KL Divergence Computation

As established in Appendix C.1.4, we assume the training and target data are independent, and that the training observations are independent and identically distributed across i . By the additivity property of KL divergence for product distributions (see, e.g., Equation 15.11 in Wainwright, 2019), we can cleanly decompose the divergence. Fix a policy π and an index $\omega \in \Omega_m$. For any alternative index $\omega' \in \Omega$ such that $\omega' \neq \omega$, the KL divergence between $\theta_{\omega', \pi}$ and $\theta_{\omega, \pi}$ decomposes as follows:

$$\text{KL}(\theta_{\omega', \pi}, \theta_{\omega, \pi}) = \text{KL}_Q + \text{KL}_P, \quad (32)$$

where the training data divergence is defined as

$$\text{KL}_P := n \mathbb{E}_{t_{\omega'}} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y_1; \mathbf{X}_1, T_1)}{\vartheta_{f_\omega}(Y_1; \mathbf{X}_1, T_1)} \right) \right], \quad (33)$$

and the target data divergence is defined as

$$\text{KL}_Q := \mathbb{E}_{Q_{\omega', \pi}} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y^Q; \mathbf{X}^Q, \pi(\mathbf{X}^Q))}{\vartheta_{f_\omega}(Y^Q; \mathbf{X}^Q, \pi(\mathbf{X}^Q))} \right) \right]. \quad (34)$$

Lemma 10 (proof provided in Appendix C.7) bounds each of KL_P and KL_Q .

Lemma 10. *The KL divergences defined in (33) and (34) are bounded from above as follows:*

$$\text{KL}_P \lesssim m \zeta n r^{2\beta+1+\alpha+d}, \quad \text{KL}_Q \lesssim r^{2\beta+1}. \quad (35)$$

Therefore, we conclude that

$$\text{KL}(\theta_{\omega', \pi}, \theta_{\omega, \pi}) \lesssim m \zeta n r^{d+1+2\beta+\alpha} + r^{1+2\beta}.$$

C.3 Parameter Space Computation: Lower bounding the Sub-optimal Gap

This section establishes that the problem instances constructed in Appendix C.1 are sufficiently separated in the parameter space by deriving a lower bound on the sub-optimality gap. Crucially, this bound is independent of π ; by design of our instances, no policy can avoid this fundamental level of sub-optimality.

We start with Lemma 11 (proof in Appendix C.8.1), which shows that the minimum optimal treatment is well-defined.

Lemma 11. For any $\omega \in \Omega_m$, the minimum optimal treatment, denoted as

$$t_\omega^\dagger(\mathbf{x}^Q) := \min \left\{ t' \in \arg \max_{t \in \mathcal{T}} f_\omega(\mathbf{x}^Q, t) \right\}, \quad (36)$$

is well-defined.

Since $t_\omega^\dagger(\mathbf{x}^Q)$ is well-defined, now we can define the sub-optimality gap of a policy under our problem instances.

Definition 16. For any policy $\pi : \mathcal{T} \rightarrow \mathcal{Y}$, and for any $\omega \in \Omega_m$, let us define

$$\text{SubOpt}(\pi, \omega) := V(\pi^*) - V(\pi) = \mathbb{E}_{\theta_{\omega, \pi}} \left[f_\omega(\mathbf{X}^Q, t_\omega^\dagger(\mathbf{X}^Q)) - f_\omega(\mathbf{X}^Q, \pi(\mathbf{X}^Q)) \right], \quad (37)$$

where we recall $\theta_{\omega, \pi}$ is the problem instance defined in Definition 15.

The primary tool for bounding the sub-optimality gap is Theorem 6. This result allows us to restrict our focus to policies that assign treatments within an $r/4$ neighborhood of the grid points $S_{\mathcal{T}, r}$, defined in (20). The proof of Theorem 6, provided in Appendix C.8.2), solely depends on the outcome distribution construction in Appendix C.1.3 and does not depend on p , q , or $\wp$.

Theorem 6 (Reduction of the policy class). For any policy $\pi : \mathcal{T} \rightarrow \mathcal{Y}$, for any $\mathbf{x}^Q \in \mathcal{X}$, if

$$\pi(\mathbf{x}^Q) \in \bigcup_{j=1}^{\lfloor 1/r \rfloor} \left([t_j^* - r/2, t_j^* + r/2] \setminus [t_j^* - r/4, t_j^* + r/4] \right), \quad (38)$$

then there exists a policy π' with

$$\pi'(\mathbf{x}^Q) \in \bigcup_{j=1}^{\lfloor 1/r \rfloor} [t_j^* - r/4, t_j^* + r/4],$$

and satisfies

$$f_\omega(\mathbf{x}^Q, \pi'(\mathbf{x}^Q)) \geq f_\omega(\mathbf{x}^Q, \pi(\mathbf{x}^Q)).$$

By Theorem 6, when lower bounding the sub-optimal gap, it suffices to consider price policies where

$$\pi(\mathbf{x}^Q) \in \bigcup_{j=1}^{\lfloor 1/r \rfloor} [t_j^* - r/4, t_j^* + r/4] \text{ for all } \mathbf{x}^Q \in \mathcal{X}. \quad (39)$$

Building upon this result and leveraging techniques from the proof of Theorem 6, we establish the following bound (proof in Appendix C.8.5).

Lemma 12. For any policy $\pi : \mathcal{T} \rightarrow \mathcal{Y}$, query point $\mathbf{x}^Q \in \mathcal{X}$, and packing indices $\omega, \omega' \in \Omega_m$, the following inequality holds:

$$\text{SubOpt}(\pi, \omega) + \text{SubOpt}(\pi, \omega') \geq C_\beta r^\beta \sum_{j=1}^m \mathbb{E}_{\theta_{\omega, \pi}} \left[\mathbb{1}\{\omega_j \neq \omega'_j\} \mathbb{1}\{\mathbf{X}^Q \in B_{r/4}(\mathbf{x}_j^*)\} \right].$$

Leveraging Lemma 12, for any policy π and for any $\omega \neq \omega' \in \Omega_m$, we have

$$\begin{aligned}
\text{SubOpt}(\pi, \omega) + \text{SubOpt}(\pi, \omega') &\stackrel{(i)}{\gtrsim} r^\beta \sum_{j=1}^m \mathbb{E}_{\theta_{\omega, \pi}} \left[\mathbb{1}\{\omega_j \neq \omega'_j\} \mathbb{1}\{\mathbf{X}^Q \in B_{r/4}(\mathbf{x}_j^*)\} \right] \\
&\stackrel{(ii)}{\gtrsim} r^{d+\beta} \rho(\omega, \omega') \\
&\stackrel{(iii)}{\geq} r^{d+\beta} \frac{m}{8} \\
&\stackrel{(iv)}{\geq} r^{d+\beta} \lfloor c_m/r \rfloor^d \geq c_m r^\beta \gtrsim r^\beta.
\end{aligned}$$

where each step uses

- (i) Lemma 12, and absorbing C_β to the constant,
- (ii) $q(\mathbf{x}) = 1$ in Definition 11,
- (iii) The cardinality of the packing in (23),
- (iv) The definition of m given in (21).

We summarize this conclusion as a lemma, because it will be reused in private minimax lower bound proof.

Lemma 13. *For any policy π and for any $\omega \neq \omega' \in \Omega_m$, we have*

$$\text{SubOpt}(\pi, \omega) + \text{SubOpt}(\pi, \omega') \gtrsim r^\beta.$$

C.4 Final Step: Application of Fano's Method

We choose the radius parameter r as:

$$r = (\zeta n)^{-\frac{1}{d+1+2\beta+\alpha}}.$$

Substituting this choice of r into our bounds, we can evaluate KL_P as follows:

$$\text{KL}_P \lesssim m \zeta n \left((\zeta n)^{-\frac{1}{d+1+2\beta+\alpha}} \right)^{d+1+2\beta+\alpha} = m \zeta n (\zeta n)^{-1} = m.$$

Next, we evaluate KL_Q :

$$\text{KL}_Q \lesssim \left((\zeta n)^{-\frac{1}{d+1+2\beta+\alpha}} \right)^{1+2\beta} = (\zeta n)^{-\frac{1+2\beta}{d+1+2\beta+\alpha}}.$$

Because $\beta \geq 0$ and $d, \alpha > 0$, the exponent is strictly negative. Therefore, for any $\zeta n \geq 1$, this term is bounded from above by 1. Since the packing size satisfies $m \geq 1$, we have:

$$\text{KL}_Q \lesssim 1 \leq m.$$

Combining two bounds, we have

$$\text{KL}(\theta_{\omega', \pi}, \theta_{\omega, \pi}) \lesssim m \zeta n r^{d+1+2\beta+\alpha} + r^{1+2\beta} \lesssim m.$$

The average KL divergence over the set of problem instances is:

$$\frac{1}{M^2} \sum_{\omega \in \Omega_m, \omega' \in \Omega_m} \text{KL}(\theta_{\omega, \pi}, \theta_{\omega', \pi}) \lesssim \frac{1}{M^2} \sum_{i \in \Omega_m, j \in \Omega_m} m = m \lesssim \log_2(M)$$

where the last inequality, up to a constant, uses $\log_2 M \geq m/8$ from (23).

Based on the small KL divergence of our selected problem instances (as shown above) and their parameter separation of r^β (as shown in (13)), Theorem 5 gives us:

$$\inf_{\pi} \sup_{\mathcal{H}_{\Omega_m}} \mathbb{E}_{\theta_{\omega}, \pi} [f(\mathbf{X}^Q, t^*(\mathbf{X}^Q)) - f(\mathbf{X}^Q, \pi(\mathbf{X}^Q))] \gtrsim r^\beta = (\zeta n)^{-\frac{\beta}{2\beta+\alpha+d+1}}.$$

This concludes the proof Theorem 1.

C.5 Proof of Theorem 5

Proof. We have for any policy π ,

$$\begin{aligned} \sup_{\theta \in \Theta} E_{D \sim P_\theta} [L(\theta, \pi(D))] &\geq \sup_{i \in [m]} E_{D \sim P_{\theta_i}} [L(\theta_i, \pi(D))] \\ &\geq \frac{1}{m} \sum_{j=1}^m E_{D \sim P_{\theta_i}} [L(\theta_i, \pi(D))]. \end{aligned}$$

Now denoting $\Psi(D) \equiv \arg \min_{i \in [m]} L(\theta_i, \pi(D))$, we get

$$L(\theta_i, \pi(D)) \geq \frac{1}{2} (L(\theta_i, \pi(D)) + L(\theta_{\Psi(D)}, \pi(D))) \geq \frac{\Delta}{2} \mathbb{I}\{\Psi(D) \neq i\},$$

where $\mathbb{I}$ is the indicator function. Thus,

$$E_{D \sim P_{\theta_i}} [L(\theta_i, \pi(D))] \geq \frac{\Delta}{2} P_{\theta_i}(\Psi \neq i),$$

taking summation over i , we get the first inequality. The second inequality follows from Proposition 15.12 and equation (15.34) of Wainwright (2019). $\square$

C.6 Proof of Lemmas Supporting Appendix C.1

C.6.1 Proof of Lemma 4

Proof. We first compute the volume of the balls. Since we are using ℓ_∞ balls in d -dimensional space, the volume of a ball $B_k(\mathbf{x}_j^*)$ with radius k (which is a hypercube of side length $2k$) is $(2k)^d$.

- Volume of the $r/2$ ball: $\text{Leb}(B_{r/2}(\mathbf{x}_j^*)) = (2 \cdot r/2)^d = r^d$
- Volume of the r ball: $\text{Leb}(B_r(\mathbf{x}_j^*)) = (2r)^d = 2^d r^d$
- Volume of the annulus: $\text{Leb}(B_r(\mathbf{x}_j^*) \setminus B_{r/2}(\mathbf{x}_j^*)) = 2^d r^d - r^d = r^d(2^d - 1)$

Let's integrate $p(\mathbf{x})$ over a single large ball $B_r(\mathbf{x}_j^*)$. We first split the integral region into the $r/2$ ball and the annulus:

$$\begin{aligned}
\int_{B_r(\mathbf{x}_j^*)} p(\mathbf{x}) d\mathbf{x} &= \int_{B_{r/2}(\mathbf{x}_j^*)} C_\alpha r^\alpha d\mathbf{x} + \int_{B_r(\mathbf{x}_j^*) \setminus B_{r/2}(\mathbf{x}_j^*)} \delta d\mathbf{x} \\
&= C_\alpha r^\alpha (r^d) + \left(\frac{2^d - C_\alpha r^\alpha}{2^d - 1} \right) \cdot r^d (2^d - 1) \\
&= C_\alpha r^{\alpha+d} + r^d (2^d - C_\alpha r^\alpha) \\
&= 2^d r^d = \text{Leb}(B_r(\mathbf{x}_j^*)). \tag{40}
\end{aligned}$$

Note that the C_α term cancels out. Consequently, the entire computation holds regardless of the value of C_α .

This makes the whole computation hold regardless of the C_α value. Now we integrate over the entire domain $\mathcal{X} = [0, 1]^d$.

$$\begin{aligned}
\int_{\mathcal{X}} p(\mathbf{x}) d\mathbf{x} &= \sum_{j=1}^m \left(\int_{B_r(\mathbf{x}_j^*)} p(\mathbf{x}) d\mathbf{x} \right) + \int_{\mathcal{X} \setminus \bigcup_{j=1}^m B_r(\mathbf{x}_j^*)} p(\mathbf{x}) d\mathbf{x} \\
&= m(2^d r^d) + 1 \cdot \text{Leb} \left(\mathcal{X} \setminus \bigcup_{j=1}^m B_r(\mathbf{x}_j^*) \right) \\
&\stackrel{(i)}{=} m2^d r^d + (1 - m2^d r^d) = 1
\end{aligned}$$

where step (i) uses the fact that by definition, $p(\mathbf{x}) = 1$ in the outside region, so its integral is just the remaining geometric volume of the space. This completes the proof of Lemma 4. $\square$

C.6.2 Proof of Lemma 5

Proof. We first establish a point-wise lower bound on the training density $p(\mathbf{x})$ relative to the uniform target density $q(\mathbf{x}) = 1$. We evaluate $p(\mathbf{x})$ across the three constructed regions:

- On the inner bump $B_{r/2}(\mathbf{x}_j^*)$: $p(\mathbf{x}) = C_\alpha r^\alpha = C_\alpha r^\alpha q(\mathbf{x})$.
- On the annulus $B_r(\mathbf{x}_j^*) \setminus B_{r/2}(\mathbf{x}_j^*)$: $p(\mathbf{x}) = \delta$. Since $C_\alpha r^\alpha \leq 1$, we have:

$$\delta = \frac{2^d - C_\alpha r^\alpha}{2^d - 1} \geq \frac{2^d - 1}{2^d - 1} = 1.$$

Thus, $p(\mathbf{x}) \geq 1 \geq C_\alpha r^\alpha q(\mathbf{x})$.

- On the outside region $\mathcal{X} \setminus \bigcup_{j=1}^m B_r(\mathbf{x}_j^*)$: $p(\mathbf{x}) = 1 \geq C_\alpha r^\alpha q(\mathbf{x})$.

Consequently, for all $\mathbf{x} \in \mathcal{X}$, we hold the global point-wise bound $p(\mathbf{x}) \geq C_\alpha r^\alpha q(\mathbf{x})$.

We now evaluate the transfer exponent condition for any query context $\mathbf{x} \in \mathcal{X}$ and bandwidth $h \in (0, 1]$, splitting the integration into two cases:

Case 1: Small bandwidth ($h \leq r$). Because the point-wise bound between densities is global, we can integrate directly over $B_h(\mathbf{x})$ without needing to restrict $\mathbf{x}$ to a specific subset

of the packing or worrying about boundary intersections:

$$\begin{aligned}
P(B_h(\mathbf{x})) &= \int_{B_h(\mathbf{x})} p(\mathbf{z}) d\mathbf{z} \\
&\geq \int_{B_h(\mathbf{x})} C_\alpha r^\alpha q(\mathbf{z}) d\mathbf{z} \\
&= C_\alpha r^\alpha \int_{B_h(\mathbf{x})} q(\mathbf{z}) d\mathbf{z} \\
&= C_\alpha r^\alpha Q(B_h(\mathbf{x})) \\
&\geq C_\alpha h^\alpha Q(B_h(\mathbf{x}))
\end{aligned}$$

where the last inequality uses the condition $h \leq r$ combined with the transfer exponent $\alpha > 0$, ensuring $r^\alpha \geq h^\alpha$.

Case 2: Large bandwidth ($h > r$). Because $p(\mathbf{y}) \geq 1$ everywhere except inside the $r/2$ -balls (where $p(\mathbf{y}) = C_\alpha r^\alpha \leq 1$), our goal is to find the maximum possible volume that the low-density $r/2$ -balls can occupy within an arbitrary query ball $B_h(\mathbf{x})$.

We first address the one-dimensional case, formalizing its general structure in Lemma 14 (proof deferred to Appendix C.6.3). The set S generalizes the context grid introduced in Definition 8. It consists of regions separated by gaps of length r , which represent the candidate centers for the balls of radius $r/2$ considered in this setting.

Lemma 14. *Let $S \subset \mathbb{R}$ be the union of intervals of length r centered at the points $2rv$ for $v \in \mathbb{Z}$. That is,*

$$S = \bigcup_{v \in \mathbb{Z}} \left[2rv - \frac{r}{2}, 2rv + \frac{r}{2} \right].$$

Then for any interval $J \subset \mathbb{R}$ of length $L > 2r$, we have

$$|J \cap S| \leq \frac{2}{3}L.$$

By applying Lemma 14 independently to each of the d dimensions, we lower-bound the integral of $P(B_h(\mathbf{x}))$ strictly using the high-density volume:

$$P(B_h(\mathbf{x})) \geq \left[1 - \left(\frac{2}{3} \right)^d \right] \text{Leb}(B_h(\mathbf{x})) \stackrel{(i)}{=} \left[1 - \left(\frac{2}{3} \right)^d \right] Q(B_h(\mathbf{x})) \stackrel{(i)}{\geq} \frac{1}{3} h^\alpha Q(B_h(\mathbf{x}))$$

where each step uses

- (i) $q(\mathbf{x})$ is a uniform distribution,
- (ii) $d \geq 1$ and $h^\alpha \leq 1$.

Since we picked $C_\alpha < 1/3$ in Definition 12, the transfer exponent condition holds here.

Combining Case 1 and Case 2 confirms that the condition $P(B_h(\mathbf{x})) \geq C_\alpha h^\alpha Q(B_h(\mathbf{x}))$ holds for all $\mathbf{x} \in \mathcal{X}$ and $h \in (0, 1]$. This completes the proof of Lemma 5. $\square$

C.6.3 Proof of Lemma 14

Proof. The set S is strictly $2r$ -periodic, consisting of alternating intervals of length r and gaps of length r . Let J be an arbitrary interval of length $L > 2r$. Geometrically, if we put the k full periods inside the interval J , then J might spill over on both ends, leaving two fractional boundary pieces on the left and right. If these the sum length of these two pieces exceed $2r$, because S is strictly $2r$ -periodic, translation invariance allows us to conceptually "cut and paste" these disjoint pieces together to create one full cycle of length $2r$ and a single, contiguous remainder interval of length y . Therefore, without loss of generality, we can partition J into just two contiguous parts:

- a main segment of length $2rk$
- a single remainder segment of length y .

Because the length of the main segment is an exact multiple of the period $2r$, it contains exactly k full periods and thus intersection with S is exactly rk . Let us call the remainder segment as I_y . The overlap $|I_y \cap S|$ is bounded by two simple geometric facts:

1. The overlap cannot be larger than the segment itself: $|I_y \cap S| \leq y$.
2. The segment I_y (length $< 2r$) sits inside some larger window of length $2r$. Since any $2r$ window contains at most r of S , the overlap inside I_y cannot exceed r : $|I_y \cap S| \leq r$.

Combining these, the remainder's contribution is strictly bounded by $\min(y, r)$. Thus, the total intersection is bounded by:

$$|J \cap S| \leq rk + \min(y, r).$$

We now verify that $rk + \min(y, r) \leq \frac{2}{3}L$ for all possible values of y . We substitute $L = 2rk + y$ and check the two cases:

- Case 1: $0 \leq y \leq r$. Here, $\min(y, r) = y$. We need to show:

$$rk + y \leq \frac{2}{3}(2rk + y),$$

which is, subtracting $\frac{2}{3}y$ and rk from both sides, equivalent to

$$y \leq rk.$$

Since $y \leq r$ (from the case condition) and $k \geq 1$ (since $L > 2r$), we know $y \leq r \leq rk$.

- Case 2: $r < y < 2r$. Here, $\min(y, r) = r$. We need to show:

$$rk + r \leq \frac{2}{3}(2rk + y),$$

which, subtracting rk from both sides, simplifies to:

$$r \leq \frac{1}{3}rk + \frac{2}{3}y.$$

Because $k \geq 1$, we know $\frac{1}{3}rk \geq \frac{1}{3}r$. Because $y > r$ (from the case condition), we know $\frac{2}{3}y > \frac{2}{3}r$. Adding these two lower bounds together yields exactly r . Thus, the right side is strictly greater than r , and the inequality always holds.

Therefore, in all possible cases, the maximum intersection satisfies the bound:

$$|J \cap S| \leq \frac{2}{3}L.$$

This completes the proof of Lemma 14. $\square$

C.6.4 Proof of Lemma 6

Proof. The proof follows the same steps as the proof in Appendix C.1 of Wang et al. (2025). We include it here to show that the transfer component assumption holds even with our weaker definition of exploration condition in Definition 2.

Pick $t^\dagger(\mathbf{x})$ as the minimum optimal price defined in (36). As shown in Lemma 11, this minimum value is guaranteed to exist. Recall that our construction of $\varphi(t|\mathbf{x})$ in equation (13) does not depend on $\mathbf{x}$. Therefore, for any $\mathbf{x} \in \mathcal{X}$ and a given $r \in (0, 1]$, we have that:

$$\mu([t^\dagger(\mathbf{x}) - r, t^\dagger(\mathbf{x}) + r] \times B_r(\mathbf{x})) = P(B_r(\mathbf{x})) 2r \min\left\{\zeta, \frac{1 - r\zeta}{1 - r}\right\} = P(B_r(\mathbf{x})) 2r\zeta,$$

where the last equality holds because

$$\zeta - \frac{1 - r\zeta}{1 - r} = \frac{\zeta(1 - r) - (1 - r\zeta)}{1 - r} = \frac{\zeta - r\zeta - 1 + r\zeta}{1 - r} = \frac{\zeta - 1}{1 - r} \leq 0.$$

As a result, the exploration coefficient defined in Definition 2 for the constructed training context-price pair distribution equals ζ . $\square$

C.6.5 Proof of Lemma 7

Proof. To prove that the function $\phi(z)$ is Hölder continuous, we need to show that there exists a constant C such that for any $z_1, z_2 \in \mathbb{R}_+$, we have $|\phi(z_1) - \phi(z_2)| \leq C|z_1 - z_2|^\beta$. We will analyze this by considering the different intervals where z_1 and z_2 can be.

Case 1: $z_1, z_2 \in [1/4, 1/2]$. In this interval, the function is defined as $\phi(z) = (2 - 4z)^\beta$. We will use the inequality $|a^\beta - b^\beta| \leq |a - b|^\beta$ for $a, b \geq 0$ and $\beta \in (0, 1]$. Let $a = 2 - 4z_1$ and $b = 2 - 4z_2$. Since $z_1, z_2 \in [1/4, 1/2]$, we have $a, b \in [0, 1]$, so the inequality applies.

$$\begin{aligned} |\phi(z_1) - \phi(z_2)| &= |(2 - 4z_1)^\beta - (2 - 4z_2)^\beta| \\ &\leq |(2 - 4z_1) - (2 - 4z_2)|^\beta \\ &= |4z_2 - 4z_1|^\beta \\ &= |4(z_2 - z_1)|^\beta \\ &= 4^\beta |z_1 - z_2|^\beta. \end{aligned}$$

This shows that on the interval $[1/4, 1/2]$, the function is Hölder continuous with constant 4^β .

Case 2: z_1, z_2 are in different intervals. We now consider the cases where z_1 and z_2 are not in the same interval. Without loss of generality, assume $z_1 < z_2$. The only nontrivial cases are when one point is in an interval where the function is constant and the other is not.

Subcase 2.1: $z_1 < 1/4$ and $z_2 \in [1/4, 1/2]$. Here, $\phi(z_1) = 1$ and $\phi(z_2) = (2 - 4z_2)^\beta$. We have

$$\begin{aligned} |\phi(z_1) - \phi(z_2)| &= 1 - (2 - 4z_2)^\beta \\ &\leq 1 - (2 - 4(z_1 + d))^\beta \\ &= 1 - (2 - 4z_1 - 4d)^\beta \end{aligned}$$

where $d = z_2 - z_1 > 0$. A simpler approach is to use the fact that the maximum value of the function's derivative is at the endpoints of the intervals. We have $|\phi(z_1) - \phi(z_2)| = 1 - (2 - 4z_2)^\beta$ and $|z_1 - z_2| = z_2 - z_1$. Since $z_1 < 1/4 \leq z_2$, we have $z_2 - z_1 > z_2 - 1/4 = (4z_2 - 1)/4$. This implies $4(z_2 - z_1) > 4z_2 - 1$. The function $f(x) = 1 - x^\beta$ on $[0, 1]$ is convex, so $1 - x^\beta \leq \beta(1 - x)$ for $\beta \leq 1$. Let's use the inequality $1 - x^\beta \leq (1 - x)^\beta$ for $x \in [0, 1]$. Let $x = 2 - 4z_2 \in [0, 1]$.

$$|\phi(z_1) - \phi(z_2)| = 1 - (2 - 4z_2)^\beta \leq (1 - (2 - 4z_2))^\beta = (4z_2 - 1)^\beta$$

Since $z_2 - z_1 \geq z_2 - 1/4$, we have $4(z_2 - z_1) \geq 4z_2 - 1$, so $4^\beta |z_2 - z_1|^\beta \geq (4z_2 - 1)^\beta$. Thus, $|\phi(z_1) - \phi(z_2)| \leq 4^\beta |z_1 - z_2|^\beta$.

Subcase 2.2: $z_1 \in [1/4, 1/2]$ and $z_2 > 1/2$. Here, $\phi(z_1) = (2 - 4z_1)^\beta$ and $\phi(z_2) = 0$. We have $|\phi(z_1) - \phi(z_2)| = (2 - 4z_1)^\beta$. Since $z_1 \in [1/4, 1/2]$ and $z_2 > 1/2$, we have $|z_1 - z_2| = z_2 - z_1$. Also, $z_2 - z_1 > 1/2 - z_1 = (2 - 4z_1)/4$. This implies $4(z_2 - z_1) > 2 - 4z_1$. Taking the β -th power of both sides, we get $4^\beta (z_2 - z_1)^\beta > (2 - 4z_1)^\beta = |\phi(z_1) - \phi(z_2)|$. Thus, $|\phi(z_1) - \phi(z_2)| \leq 4^\beta |z_1 - z_2|^\beta$.

Subcase 2.3: $z_1 < 1/4$ and $z_2 > 1/2$. In this case, $|\phi(z_1) - \phi(z_2)| = |1 - 0| = 1$. Also, since $z_1 < 1/4$ and $z_2 > 1/2$, we have $|z_1 - z_2| = z_2 - z_1 > 1/2 - 1/4 = 1/4$. Taking the β -th power, $|z_1 - z_2|^\beta > (1/4)^\beta$. Multiplying by 4^β , we get $4^\beta |z_1 - z_2|^\beta > 4^\beta (1/4)^\beta = 1$. So, $|\phi(z_1) - \phi(z_2)| = 1 < 4^\beta |z_1 - z_2|^\beta$.

In all cases, the inequality $|\phi(z_1) - \phi(z_2)| \leq 4^\beta |z_1 - z_2|^\beta$ holds. Therefore, $\phi(z)$ is Hölder continuous with exponent β and Hölder constant 4^β . $\square$

C.6.6 Proof of Lemma 8

Proof. By the construction in (29), it suffices to show that the function $\varphi_\beta : \mathcal{X} \times \mathcal{T} \rightarrow [0, 1/4]$ satisfies assumptions on conditional outcome in Assumption 1 with respect to the Hölder constant $C_H > 0$. For any $(\mathbf{x}, t), (\mathbf{x}', t') \in \mathcal{X} \times \mathcal{T}$, we have that

$$|\varphi_\beta(\mathbf{x}, t) - \varphi_\beta(\mathbf{x}', t')| = C_{\varphi_\beta} r \left| \phi\left(\max\{\|\mathbf{x}\|_\infty, |t|\}/r\right) - \phi\left(\max\{\|\mathbf{x}'\|_\infty, |t'|\}/r\right) \right| \quad (41)$$

Since ϕ is a Hölder continuous function with Hölder constant 4^β , it follows that

$$\begin{aligned} |\varphi_\beta(\mathbf{x}, t) - \varphi_\beta(\mathbf{x}', t')| &\leq 4^\beta C_{\varphi_\beta} r \left| \max\{\|\mathbf{x}\|_\infty, |t|\}/r - \max\{\|\mathbf{x}'\|_\infty, |t'|\}/r \right| \\ &= 4^\beta C_{\varphi_\beta} \left| \max\{\|\mathbf{x}\|_\infty, |t|\} - \max\{\|\mathbf{x}'\|_\infty, |t'|\} \right| \\ &\stackrel{(i)}{\leq} 4^\beta C_{\varphi_\beta} \max\left\{ \left| \|\mathbf{x}\|_\infty - \|\mathbf{x}'\|_\infty \right|, \left| |t| - |t'| \right| \right\} \\ &\stackrel{(ii)}{\leq} 4^\beta C_{\varphi_\beta} \max\left\{ \|\mathbf{x} - \mathbf{x}'\|_\infty, |t - t'| \right\} \\ &\leq C_H \max\left\{ \|\mathbf{x} - \mathbf{x}'\|_\infty, |t - t'| \right\}, \end{aligned}$$

where step (i) uses a basic inequality $|\max(a, b) - \max(c, d)| \leq \max(|a - c|, |b - d|)$, step (ii) uses the reverse triangle inequality, and step (iii) holds since $C_{\varphi_\beta} = (C_H/4^\beta) \wedge (1/4^\beta)$. Thus, for any $\omega \in \Omega_m$, the outcome function f_ω satisfies assumptions on conditional outcome in Assumption 1 with respect to the Hölder constant $C_H > 0$. $\square$

C.7 Proof of Lemma 10

Proof. We start by bounding KL_P :

$$\begin{aligned}
\text{KL}_P &= n \mathbb{E}_{t_{\omega'}} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y_1; \mathbf{X}_1, T_1)}{\vartheta_{f_\omega}(Y_1; \mathbf{X}_1, T_1)} \right) \right] \\
&= n \int_{[0,1]^d} \mathbb{E} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y_1; \mathbf{x}, T_1)}{\vartheta_{f_\omega}(Y_1; \mathbf{x}, T_1)} \right) \middle| \mathbf{X}_1 = \mathbf{x} \right] p(\mathbf{x}) d\mathbf{x} \\
&\stackrel{(i)}{=} n \int_{\bigcup_{j=1}^m B_{r/2}(\mathbf{x}_j^*)} \mathbb{E} \left[\mathbb{1}\{T_1 \in [\tilde{t} - r/2, \tilde{t} + r/2]\} \log \left(\frac{\vartheta_{f_{\omega'}}(Y_1; \mathbf{x}, t_1)}{\vartheta_{f_\omega}(Y_1; \mathbf{x}, t_1)} \right) \middle| \mathbf{X}_1 = \mathbf{x} \right] p(\mathbf{x}) d\mathbf{x} \\
&= n \int_{\bigcup_{j=1}^m B_{r/2}(\mathbf{x}_j^*)} \int_{\tilde{t}-r/2}^{\tilde{t}+r/2} \mathbb{E}_{\omega'} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y_1; \mathbf{x}, T_1)}{\vartheta_{f_\omega}(Y_1; \mathbf{x}, T_1)} \right) \middle| \mathbf{X}_1 = \mathbf{x}, T_1 = t \right] p(\mathbf{x}) \varphi(t | \mathbf{x}) dt d\mathbf{x} \\
&\stackrel{(ii)}{\lesssim} n \sum_{j \in [m]: \omega'_j \neq \omega_j} P(B_{r/2}(\mathbf{x}_j^*)) r \zeta \text{kl}(1/2 + C_\beta r^\beta \parallel 1/2 - C_\beta r^\beta) \\
&\stackrel{(iii)}{\lesssim} n r \zeta (C_\beta r^\beta)^2 \sum_{j \in [m]: \omega'_j \neq \omega_j} P(B_{r/2}(\mathbf{x}_j^*)) \\
&\stackrel{(iv)}{\leq} C_\alpha n \zeta m r^{2\beta+1+\alpha+d} \\
&\stackrel{(v)}{\lesssim} \zeta m n r^{2\beta+1+\alpha+d},
\end{aligned}$$

where each step uses:

- (i) Lemma 9,
- (ii) $\varphi(t | \mathbf{x}) = \zeta$ for $t \in [\tilde{t} - r/2, \tilde{t} + r/2]$, regardless of $\mathbf{x}$, as defined in (13), and the Bernoulli construction of the outcome distribution and outcome function given in (29), and by Lemma 7 and Definition 14, these Bernoulli success probabilities evaluate exactly to $1/2 + C_\beta r^\beta$ and $1/2 - C_\beta r^\beta$ inside any $r/4$ -ball in $\mathcal{X}$. Furthermore, on the annulus between the $r/4$ -ball and the $r/2$ -ball, the success probabilities are identically $1/2$ for all $\mathbf{x}$.
- (iii) Lemma 3,
- (iv) Absorb C_β^2 into the constant, and $p(\mathbf{x})$ is uniformly equal to $C_\alpha r^\alpha$ across the entirety of the $r/2$ -ball. Consequently, the local probability mass evaluates exactly to the geometric volume multiplied by the density: $P(B_{r/2}(\mathbf{x}_j^*)) = C_\alpha r^{\alpha+d}$.

Next, we switch our gears to bounding KL_Q from above, using similar arguments as above:

$$\begin{aligned}
& \mathbb{E}_{Q_{\omega', \pi}} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y^Q; \mathbf{X}^Q, \pi(\mathbf{X}^Q))}{\vartheta_{f_{\omega}}(Y^Q; \mathbf{X}^Q, \pi(\mathbf{X}^Q))} \right) \right] \\
&= \int_{\mathcal{X}} \mathbb{E} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y^Q; \mathbf{x}, \pi(\mathbf{x}))}{\vartheta_{f_{\omega}}(Y^Q; \mathbf{x}, \pi(\mathbf{x}))} \right) \middle| \mathbf{X}^Q = \mathbf{x} \right] q(\mathbf{x}) d\mathbf{x} \\
&\stackrel{(i)}{=} \int_{\bigcup_{j=1}^k B_{r/2}(\mathbf{x}_j^*)} \mathbb{E} \left[\mathbb{1}\{\pi(\mathbf{x}) \in [\tilde{t} - r/2, \tilde{t} + r/2]\} \log \left(\frac{\vartheta_{f_{\omega'}}(Y^Q; \mathbf{x}, \pi(\mathbf{x}))}{\vartheta_{f_{\omega}}(Y^Q; \mathbf{x}, \pi(\mathbf{x}))} \right) \middle| \mathbf{X}^Q = \mathbf{x} \right] q(\mathbf{x}) d\mathbf{x} \\
&\stackrel{(ii)}{=} \sum_{j \in [k]: \omega'_j \neq \omega_j} \int_{B_{r/2}(\mathbf{x}_j^*)} I_{r, \pi}^Q(\tilde{t}) \mathbb{E} \left[\log \left(\frac{\vartheta_{f_{\omega'}}(Y^Q; \mathbf{x}, \pi(\mathbf{x}))}{\vartheta_{f_{\omega}}(Y^Q; \mathbf{x}, \pi(\mathbf{x}))} \right) \middle| \mathbf{X}^Q = \mathbf{x} \right] q(\mathbf{x}) d\mathbf{x} \\
&\stackrel{(iii)}{\leq} \sum_{j \in [k]: \omega'_j \neq \omega_j} \int_{B_{r/2}(\mathbf{x}_j^*)} I_{r, \pi}^Q(\tilde{t}) \text{kl}(1/2 + C_{\beta} r^{\beta} \parallel 1/2 - C_{\beta} r^{\beta}) q(\mathbf{x}) d\mathbf{x} \\
&= \text{kl}(1/2 + C_{\beta} r^{\beta} \parallel 1/2 - C_{\beta} r^{\beta}) \sum_{j \in [k]: \omega'_j \neq \omega_j} \int_{B_{r/2}(\mathbf{x}_j^*)} I_{r, \pi}^Q(\tilde{t}) q(\mathbf{x}) d\mathbf{x} \\
&\stackrel{(iv)}{\leq} \text{kl}(1/2 + C_{\beta} r^{\beta} \parallel 1/2 - C_{\beta} r^{\beta}) \int_{\mathcal{X}} I_{r, \pi}^Q(\tilde{t}) q(\mathbf{x}) d\mathbf{x} \\
&= \text{kl}(1/2 + C_{\beta} r^{\beta} \parallel 1/2 - C_{\beta} r^{\beta}) \mathbb{E}_Q [I_{r, \pi}^Q(\tilde{t})] \\
&\stackrel{(v)}{\lesssim} (C_{\beta} r^{\beta})^2 \frac{1}{\lfloor 1/r \rfloor} \\
&\stackrel{(vi)}{\lesssim} r^{2\beta+1}
\end{aligned}$$

where each step uses:

- (i) Lemma 9, which ensures the log-likelihood ratio is identically zero outside the active spatial bumps and the active price interval.
- (ii) Recalling the definition from (27), $I_{r, \pi}^Q(\tilde{t}) = \mathbb{1}\{\pi(\mathbf{X}^Q) \in [\tilde{t} - r/2, \tilde{t} + r/2]\}$ is a transformation of $\mathbf{X}^Q$. Conditioned on $\mathbf{X}^Q = \mathbf{x}$, this indicator is deterministic and can be factored out of the expectation.
- (iii) Conditioning on $\mathbf{x}$ fixes $\pi(\mathbf{x})$, completely determining the outcome distributions as Bernoulli with fixed success probabilities. Thus, the inner expectation evaluates exactly to the KL divergence between two Bernoulli distributions. By Lemma 7 and Definition 14, these success probabilities are exactly $1/2 \pm C_{\beta} r^{\beta}$ everywhere inside the $r/4$ -balls in $\mathcal{X}$, and are identically equal to $1/2$ for all $\mathbf{x}$ within the intermediate annuli between the $r/4$ and $r/2$ radii. (We upper-bound the integral over the entire $r/2$ -ball by the maximum divergence).
- (iv) Since the integrands are non-negative, the sum of disjoint integrals over the localized $r/2$ -balls is strictly upper-bounded by extending the integral over the entire domain $\mathcal{X}$, which evaluates to the global expectation.

- (v) Lemma 3 to bound the Bernoulli KL divergence, alongside the pigeonhole principle from (28) which guarantees $\mathbb{E}_Q[I_{r,\pi}^Q(\tilde{t})] \leq 1/\lfloor 1/r \rfloor$.
- (vi) Absorbing C_β^2 into the constant and using the inequality $1/\lfloor 1/r \rfloor \lesssim r$.

This completes the proof of Lemma 10. $\square$

C.8 Proof of Lemmas Supporting Appendix C.3

C.8.1 Proof of Lemma 11

Proof. For any $\omega \in \Omega_m$, by our construction, the outcome function $f_\omega(\mathbf{x}, t)$ is continuous with respect to t for any fixed $\mathbf{x} \in \mathcal{X}$. Since the domain for the treatment, $\mathcal{T} = [0, 1]$, is a compact set, the Extreme Value Theorem guarantees that a maximum value, M , is attained. The set of all treatments that achieve this maximum value, known as the arg max set, can be expressed as the preimage of the closed set $\{M\}$ under the continuous function $f_\omega(\mathbf{x}, \cdot)$, i.e., $\{t \in \mathcal{T} \mid f_\omega(\mathbf{x}, t) = M\}$. Since the preimage of a closed set by a continuous function is still closed, arg max set is a closed subset of the compact domain $\mathcal{T}$, and thus is also compact. Finally, because the arg max set is a non-empty compact set of real numbers, it is guaranteed to contain a minimum element. Therefore, the minimal optimal price, $t_\omega^\dagger(\mathbf{x})$, is well-defined. This completes the proof of Lemma 11. $\square$

C.8.2 Proof of Theorem 6

Proof. We start with Lemma 15, which states that when the context is sufficiently close to one of the grid points, the minimum optimal treatments corresponding to different bump directions are simply specified. The proof of Lemma 15, provided in Appendix C.8.3, solely depends on the outcome distribution construction in Appendix C.1.3 and does not depend on p , q , or $\wp$.

Lemma 15 (Optimal treatment near context grid point). *Given $\omega \in \Omega_m$ and $\mathbf{x} \in \mathcal{X}$, if there exists $j \in [m]$ such that $\mathbf{x} \in B_{r/4}(\mathbf{x}_j^*)$:*

1. *If $\omega_j = -1$, then $t_\omega^\dagger(\mathbf{x}) = 0$.*
2. *If $\omega_j = 1$, then $t_\omega^\dagger(\mathbf{x}) = \tilde{t} - r/4$.*

For the next lemma, we define the function $h : \mathcal{T} \rightarrow \{0, 1\}$ to characterize indicate whether the treatment t falls outside the flat region defined in Lemma 9:

$$h(t) := \mathbb{1} \left\{ t \in \left(\tilde{t} - \frac{r}{2}, \tilde{t} + \frac{r}{2} \right] \right\}.$$

The following is a corollary of Lemma 15.

Corollary 1. *Fix $\mathbf{x} \in \mathcal{X}$ and two distinct $\omega, \omega' \in \Omega_m$. If there exists an index $j \in [m]$ such that $\omega_j \neq \omega'_j$ and $\mathbf{x} \in B_{r/4}(\mathbf{x}_j^*)$, then exactly one of $t_\omega^\dagger(\mathbf{x})$ and $t_{\omega'}^*(\mathbf{x})$ lies in $\left(\tilde{t} - \frac{r}{2}, \tilde{t} + \frac{r}{2} \right]$, while the other lies outside it. More formally,*

$$h(t_\omega^\dagger(\mathbf{x})) \neq h(t_{\omega'}^*(\mathbf{x})).$$

Next, Lemma 16 states that for contexts and treatments sufficiently close to grid points, the sub-optimality gap is at most r^β , up to a constant. Moreover, the sub-optimality gap is zero if only one of the optimal treatment and the chosen treatment t fall into $(\tilde{t} - \frac{r}{2}, \tilde{t} + \frac{r}{2}]$. The proof of Lemma 16, provided in Appendix C.8.4, solely depends on the outcome distribution construction in Appendix C.1.3 and does not depend on p , q , or φ .

Lemma 16. *For any $\omega \in \Omega_m$, $\mathbf{x} \in \bigcup_{j=1}^m B_{r/4}(\mathbf{x}_j^*)$, and $t \in \bigcup_{j=1}^{\lfloor 1/r \rfloor} [t_j^* - r/4, t_j^* + r/4]$, the following holds:*

$$f_\omega(\mathbf{x}, t_\omega^*(\mathbf{x})) - f_\omega(\mathbf{x}, t) = C_\beta r^\beta \mathbb{1}\{h(t_\omega^*(\mathbf{x})) \neq h(t)\}.$$

Now we are ready to prove the main theorem. Let us assume that $\pi(\mathbf{x}) \in [t_j^* - r/2, t_j^* + r/2] \setminus [t_j^* - r/4, t_j^* + r/4]$ for some $j \in [\lfloor 1/r \rfloor]$. We will analyze this situation by considering two primary cases based on the location of $\mathbf{x}$.

Case 1: $\mathbf{x} \in [0, 1]^d \setminus \bigcup_{l=1}^m B_{r/2}(x_l^*)$. This is an easy case. If $\mathbf{x}$ lies outside the union of the balls $B_{r/2}(\mathbf{x}_j^*)$, then, in accordance with Lemma 9, the outcome function $f_\omega(\mathbf{x}, t)$ evaluates to $1/2$ for all $\omega \in \Omega_m$ and for any treatment $t \in [0, 1]$. This implies that the specific treatment chosen has no influence on the resulting outcome. Therefore, we are at liberty to select $\pi'(\mathbf{x})$ such that it falls within the desired interval:

$$\pi'(\mathbf{x}) \in [t_j^* - r/4, t_j^* + r/4].$$

Case 2: $\mathbf{x} \in \bigcup_{l=1}^m B_{r/2}(x_l^*)$. This is a non-trivial case. We consider the subcases of $\omega_j = 1$ and $\omega_j = -1$.

Case 2.1: $\omega_j = 1$. We proceed by examining two distinct sub-scenarios:

- **If $t_j^* \neq \tilde{t}$:** Recall that the treatment grids $\mathcal{S}_{\mathcal{T}, r}$, defined in (20), have r -spacing. Thus for any treatment $t \in [t_j^* - r/2, t_j^* + r/2]$, we have $|\tilde{t} - t| \geq r/2$. Thus by Lemma 9, the outcome function value remains the same for any $t \in [t_j^* - r/2, t_j^* + r/2]$. Consequently, by setting $\pi'(\mathbf{x}) \in [t_j^* - r/4, t_j^* + r/4]$, the outcome associated with π' will be identical to that of π .
- **If $t_j^* = \tilde{t}$:** The positive bump of the outcome function, formally represented by φ_β as defined in (26) and (29), is a non-decreasing function of the distance between t and $t_j^* = \tilde{t}$. Thus, if we choose

$$\pi'(\mathbf{x}) \in [t_j^* - r/4, t_j^* + r/4],$$

this selection positions $\pi'(\mathbf{x})$ closer to $\tilde{t}$ than $\pi(\mathbf{x})$. Consequently, the outcome function value for π' will be greater than that for π .

Case 2.2: $\omega_j = -1$ We proceed by examining two distinct sub-scenarios:

- **If $t_j^* \neq \tilde{t}$:** Consistent with the reasoning in Case 2.1, we can safely choose $\pi'(\mathbf{x}) \in [t_j^* - r/4, t_j^* + r/4]$.
- **If $t_j^* = \tilde{t}$:** The negative bump of the outcome function, formally represented by $-\varphi_\beta$ as defined in (26) and (29), is a non-decreasing function of the distance between p and t_j^* . Thus, if we opt for

$$\pi'(\mathbf{x}) \in [t_k^* - r/4, t_k^* + r/4], \quad \text{for some } k \neq j,$$

this selection places $\pi'(\mathbf{x})$ further from $\tilde{t}$ than $\pi(\mathbf{x})$. As a result, the outcome function value for π' will be greater than that for π . This completes the proof of Theorem 6. $\square$

C.8.3 Proof of Lemma 15

Proof. Since $\mathbf{x} \in B_{r/4}(\mathbf{x}_j^*)$ for some $j \in [m]$, by definition of the L_∞ -norm ball, we have $\|\mathbf{x} - \mathbf{x}_j^*\|_\infty < r/4$. Due to the $2r$ -spacing of the grid points $S_{\mathcal{X},r}$ and the localization of the outcome function's bumps within r -balls, the outcome function simplifies to a single bump. We now analyze the two cases for ω_j :

Case 1: $\omega_j = -1$. The outcome function becomes $f_\omega(\mathbf{x}, t) = \frac{1}{2} - \varphi_\beta(\mathbf{x} - \mathbf{x}_j^*, t - \tilde{t})$. To maximize this function, we must minimize the non-negative term $\varphi_\beta(\mathbf{x} - \mathbf{x}_j^*, t - \tilde{t})$. The function φ_β , defined in (26) is zero when its argument $z := \frac{1}{r} \max\{\|\mathbf{x} - \mathbf{x}_j^*\|_\infty, |t - \tilde{t}|\}$ satisfies $z \geq 1/2$. This is equivalent to $\max\{\|\mathbf{x} - \mathbf{x}_j^*\|_\infty, |t - \tilde{t}|\} \geq r/2$. Given that $\|\mathbf{x} - \mathbf{x}_j^*\|_\infty < r/4$, this condition simplifies to requiring $|t - \tilde{t}| \geq r/2$. We need to find the lowest treatment t that satisfies this condition. As $\tilde{t}$ belongs to the grid point set $S_{\mathcal{T},r}$, which starts from $r/2$, it holds that $\tilde{t} \geq r/2$. Consequently $p = 0$ is the smallest $t \in \mathcal{T}$ that satisfies $|t - \tilde{t}| \geq r/2$, since $|0 - \tilde{t}| = \tilde{t} \geq r/2$. Therefore 0 is the optimal price.

Case 2: $\omega_j = 1$. The outcome function becomes $f_\omega(\mathbf{x}, t) = \frac{1}{2} + \varphi_\beta(\mathbf{x} - \mathbf{x}_j^*, t - \tilde{t})$. To maximize $f_\omega(\mathbf{x}, t)$, we must maximize $\varphi_\beta(\mathbf{x} - \mathbf{x}_j^*, t - \tilde{t})$. The maximum value of φ_β is achieved when its argument $z = \frac{1}{r} \max\{\|\mathbf{x} - \mathbf{x}_j^*\|_\infty, |t - \tilde{t}|\}$ satisfies $0 \leq z \leq 1/4$. So, we need $\max\{\|\mathbf{x} - \mathbf{x}_j^*\|_\infty, |t - \tilde{t}|\} \leq r/4$. Since $\|\mathbf{x} - \mathbf{x}_j^*\|_\infty < r/4$, the condition simplifies to $|t - \tilde{t}| \leq r/4$. The smallest such p is $\tilde{t} - r/4$, if it is nonnegative. Since $\tilde{t} \geq r/2$, we have $\tilde{t} - r/4 \geq r/4 \geq 0$, ensuring this is a valid non-negative price. Therefore $\tilde{t} - r/4$ is the optimal price. This completes the proof of Lemma 15. $\square$

C.8.4 Proof of Lemma 16

Proof. We first note that $\mathbf{x} \in B_{r/4}(\mathbf{x}_j^*)$ for some $j \in [m]$. If $\omega_j = -1$, we have $t_\omega^\dagger(\mathbf{x}) = 0$ by Lemma 15. Since $0 \notin (\tilde{t} - r/2, \tilde{t} + r/2]$, we have $h(t_\omega^\dagger(\mathbf{x})) = 0$. Therefore,

$$\begin{aligned} f_\omega(\mathbf{x}, t_\omega^\dagger(\mathbf{x})) - f_\omega(\mathbf{x}, t) &\stackrel{(i)}{=} C_\phi r^\beta \mathbb{1}\{|t - \tilde{t}| \leq r/2\} \\ &= C_\phi r^\beta h(t) \\ &= C_\phi r^\beta \mathbb{1}\{h(t_\omega^\dagger(\mathbf{x})) \neq h(t)\}. \end{aligned}$$

where step (i) uses $t \in \bigcup_{j=1}^{\lfloor 1/r \rfloor} [t_j^* - r/4, t_j^* + r/4]$.

Next, if $\omega_j = 1$, we have $t_\omega^\dagger(\mathbf{x}) = \tilde{t} - r/4$ by Lemma 15, and $h(t_\omega^\dagger(\mathbf{x})) = 1$. Hence,

$$\begin{aligned} f_\omega(\mathbf{x}, t_\omega^\dagger(\mathbf{x})) - f_\omega(\mathbf{x}, t) &= C_\phi r^\beta [1 - \mathbb{1}\{|t - \tilde{t}| \leq r/2\}] \\ &= C_\phi r^\beta [1 - h(t)] \\ &= C_\phi r^\beta \mathbb{1}\{h(t_\omega^\dagger(\mathbf{x})) \neq h(t)\}. \end{aligned}$$

This completes the proof of Lemma 16. $\square$

C.8.5 Proof of Lemma 12

Proof. By taking expectation on the result of Lemma 16, we have that

$$\begin{aligned} \text{SubOpt}(\pi, \omega) &\geq C_\beta r^\beta \mathbb{E}_{\theta_{\omega, \pi}} \left[\mathbb{1}\{h(t_\omega^*(\mathbf{X}^Q)) \neq h(\pi(\mathbf{X}^Q))\} \mathbb{1}\{\mathbf{X}^Q \in \bigcup_{j=1}^m B_{r/4}(\mathbf{x}_j^*)\} \right] \\ &= C_\beta r^\beta \sum_{j=1}^m \mathbb{E}_{\theta_{\omega, \pi}} \left[\mathbb{1}\{h(t_\omega^*(\mathbf{X}^Q)) \neq h(\pi(\mathbf{X}^Q))\} \mathbb{1}\{\mathbf{X}^Q \in B_{r/4}(\mathbf{x}_j^*)\} \right]. \end{aligned}$$

If $\mathbf{X}^Q \notin \bigcup_{j=1}^m B_{r/4}(\mathbf{x}_j^*)$, the right-hand side evaluates to zero. Because $\text{SubOpt}(\pi, \omega)$ is strictly non-negative, the resulting bound is trivial yet valid.

Leveraging this inequality, for any policy π satisfying (39) and for any $\omega \neq \omega' \in \Omega_m$, we have

$$\begin{aligned} &\text{SubOpt}(\pi, \omega) + \text{SubOpt}(\pi, Q, \omega') \\ &= C_\beta r^\beta \sum_{j=1}^m \mathbb{E}_{\theta_{\omega, \pi}} \left[\left(\mathbb{1}\{h(t_\omega^*(\mathbf{X}^Q)) \neq h(\pi(\mathbf{X}^Q))\} + \mathbb{1}\{h(t_{\omega'}^*(\mathbf{X}^Q)) \neq h(\pi(\mathbf{X}^Q))\} \right) \mathbb{1}\{\mathbf{X}^Q \in B_{r/4}(\mathbf{x}_j^*)\} \right] \\ &\geq C_\beta r^\beta \sum_{j=1}^m \mathbb{E}_{\theta_{\omega, \pi}} \left[\mathbb{1}\{h(t_\omega^*(\mathbf{X}^Q)) \neq h(t_{\omega'}^*(\mathbf{X}^Q))\} \mathbb{1}\{\mathbf{X}^Q \in B_{r/4}(\mathbf{x}_j^*)\} \right] \\ &\stackrel{(i)}{\geq} C_\beta r^\beta \sum_{j=1}^m \mathbb{E}_{\theta_{\omega, \pi}} \left[\mathbb{1}\{\omega_j \neq \omega'_j\} \mathbb{1}\{\mathbf{X}^Q \in B_{r/4}(\mathbf{x}_j^*)\} \right], \end{aligned}$$

where the last inequality uses Corollary 1. This completes the Proof of Lemma 12. $\square$

D Proof of Theorem 3

This section proves the private minimax lower bound presented in Theorem 3. The proof largely follows the Fano-type argument established for the non-private setting in Appendix C. The primary difference lies in the application of a distribution distance contraction (Lemma 17), which upper bounds the KL divergence by the total variation (TV) distance multiplied by the privacy parameter. Consequently, we must also compute the TV distance between problem instances, whereas the non-private proof required only the KL divergence.

D.1 Construction of Problem Instances

We adopt the problem instance construction from Appendix D.1, with the addition of a privacy mechanism. Specifically, we retain the following components:

- Packing index set Ω_m (Definition 10)
- Target context density $q(\cdot)$ (Definition 11)
- training context density $p(\cdot)$ (Definition 12)
- Offline logging policy propensity $\wp(\cdot \mid \mathbf{x}_i)$ (Definition 13)
- Outcome function $f_\omega(\mathbf{x}, t)$ (Definition 14)

For each packing index $\omega \in \Omega_m$, the raw data generation proceeds exactly as described in Appendix C.1.4. The key difference in this setting is that each training data triplet is

subsequently passed through an ϵ -LDP mechanism M . This yields the privatized training dataset:

$$\mathcal{D}_{\omega,\epsilon} = \left\{ M(\mathbf{X}_i, T_i, Y_i) \right\}_{i=1}^n,$$

which has joint distributions denoted by $\theta_{\omega,\epsilon,i}$. We assume independence across $\theta_{\omega,\epsilon,i}$ over $i = 1, \dots, n$. The target data generation remains unchanged, with its joint distribution still denoted by $Q_{\omega,\pi}$. Finally, we assume the privatized training data and the target data are independent. The full problem instance, which is a product distribution $\theta_{\omega,\epsilon,1} \times \dots, \theta_{\omega,\epsilon,n} \times Q_{\omega,\pi}$, is denoted as $\theta_{\omega,\epsilon,\pi}$.

D.2 Computation in Distribution Space: Upper Bounding the KL Divergence

This section bounds the KL divergence

$$\text{KL}(\theta_{\omega,\epsilon,\pi} \parallel \theta_{\omega',\epsilon,\pi})$$

from above. First off, as in (32), we use the additivity property of KL divergence for product distributions (see, e.g., Equation 15.11 in Wainwright, 2019) to cleanly decompose the $\text{KL}(\theta_{\omega,\epsilon,\pi} \parallel \theta_{\omega',\epsilon,\pi})$ into:

$$\text{KL}(\theta_{\omega,\epsilon,\pi} \parallel \theta_{\omega',\epsilon,\pi}) = n\text{KL}(\theta_{\omega,\epsilon,1} \parallel \theta_{\omega',\epsilon,1}) + \text{KL}_Q$$

where KL_Q is defined in (34) and was bounded in (35) as

$$\text{KL}_Q \lesssim mr^{d+1+2\beta}.$$

Therefore, the main task in bounding $\text{KL}(\theta_{\omega,\epsilon,1} \parallel \theta_{\omega',\epsilon,1})$. Given a non-private distribution $\theta_{\omega,1}$, a conditional distribution M transforms such a pair of distributions into a new pair $\theta_{\omega,\epsilon,1}$ via the marginalization operation:

$$\theta_{\omega,\epsilon,1}(S) = \int M(S \mid \mathbf{x}_i, t_i, y_i) d\theta_{\omega,1}(\mathbf{x}_i, t_i, y_i).$$

Intuitively, when the conditional distribution M is ϵ -locally differentially private, the output distributions $(\theta_{\omega,\epsilon,1}, \theta_{\omega',\epsilon,1})$ should be closer together because of the privacy noise. The following theorem makes this intuition precise:

Lemma 17 (Adapted from Corollary 3 of Duchi et al., 2018). *For $\epsilon \in [0, 1]$, let M be a conditional distribution that guarantees ϵ -LDP. Then, the induced marginals satisfy the bound:*

$$\text{KL}(\theta_{\omega,\epsilon,1} \parallel \theta_{\omega',\epsilon,1}) + \text{KL}(\theta_{\omega',\epsilon,1} \parallel \theta_{\omega,\epsilon,1}) \leq 12\epsilon^2 \|\theta_{\omega,1} - \theta_{\omega',1}\|_{\text{TV}}^2. \quad (42)$$

The following lemma (proof in Appendix D.4) computes the TV distance between non-private training distribution instances.

Lemma 18. *For $\omega \neq \omega'$ that differ in exactly one element, the TV distance between the training joint distributions of the context, treatment, and outcome is*

$$\|P_\omega - P_{\omega'}\|_{\text{TV}} \lesssim \zeta r^{\alpha+d+\beta+1}.$$

For any $\omega \neq \omega' \in \Omega_m$, let ω'' be a packing index that differs with ω by exactly one entry. The KL divergence between private training distributions is bounded as follows:

$$\begin{aligned}
n \text{KL}(\theta_{\omega, \epsilon, 1} \parallel \theta_{\omega', \epsilon, 1}) &\stackrel{(i)}{\leq} 12n\epsilon^2 \|\theta_{\omega, 1} - \theta_{\omega', 1}\|_{\text{TV}}^2 \\
&\stackrel{(ii)}{\leq} 24n\epsilon^2 \rho(\omega, \omega') \|\theta_{\omega, 1} - \theta_{\omega'', 1}\|_{\text{TV}}^2 \\
&\stackrel{(iii)}{=} 24n\epsilon^2 \rho(\omega, \omega') (C_{\text{privateTV}} \zeta r^{\alpha+d+\beta+1})^2 \\
&\stackrel{(iv)}{\leq} 8n\epsilon^2 m (C_{\text{privateTV}} \zeta r^{\alpha+d+\beta+1})^2 \\
&= C_{\text{privateKL}} n m \zeta^2 r^{2(\alpha+d+\beta+1)}
\end{aligned}$$

where each step uses

- (i) Lemma 17,
- (ii) Triangle inequality of total variation distance, and a basic inequality $(a + b + c \dots)^2 \leq 2(a^2 + b^2 + \dots)$
- (iii) Lemma 18
- (iv) Inequality (23) in Definition 10

Note that the usage of Lemma 17 made KL to scale with **squared** mass $r^{2(\alpha+d+1)}$, unlike the non-private setting where the KL scales with the probability mass $r^{\alpha+d+1}$, effectively doubling the dimension dependence.

In summary, we have

$$\text{KL}(\theta_{\omega, \epsilon, \pi} \parallel \theta_{\omega', \epsilon, \pi}) = n \text{KL}(\theta_{\omega, \epsilon, 1} \parallel \theta_{\omega', \epsilon, 1}) + \text{KL}_Q \lesssim m r^{d+1+2\beta} + n m \epsilon^2 \zeta^2 r^{2(\alpha+d+\beta+1)}. \quad (43)$$

D.3 Application of Fano's Method

If we set

$$r = \left(n \epsilon^2 \zeta^2 \right)^{-\frac{1}{2(\alpha+d+1+\beta)}}, \quad (44)$$

we have, from (43):

$$\text{KL}(\theta_{\omega, \epsilon, \pi} \parallel \theta_{\omega', \epsilon, \pi}) \lesssim m \left(n \epsilon^2 \zeta^2 \right)^{-\frac{d+1+2\beta}{2(\alpha+d+1+\beta)}} + m \lesssim m,$$

where the last inequality follows from assumption $n \epsilon^2 \zeta^2 > 1$.

The average KL divergence over the set of problem instances is:

$$\frac{1}{M^2} \sum_{j \in \Omega_m, k \in \Omega_m} \text{KL}(\theta_{\omega_j, \epsilon, \pi}, \theta_{\omega_k, \epsilon, \pi}) \lesssim \frac{1}{M^2} \sum_{j \in \Omega_m, k \in \Omega_m} m = m \lesssim \log_2(M)$$

where the last inequality, up to a constant, uses $\log_2 M \geq m/8$ from (23).

Based on the small KL divergence of our selected problem instances (as shown above) and their parameter separation of $(\zeta n)^{-\beta/(d+1+2\beta+\alpha)}$ (as shown in (13)), Theorem 5 gives us:

$$\inf_{\pi} \sup_{t_{\omega} \in \mathcal{H}_{\Omega_m}} \mathbb{E} \left[f(\mathbf{X}^Q, t^*(\mathbf{X}^Q)) - f(\mathbf{X}^Q, \pi(\mathbf{X}^Q)) \right] \gtrsim r^{\beta} = \left(n \epsilon^2 \zeta^2 \right)^{-\frac{\beta}{2(\alpha+d+1+\beta)}}.$$

This concludes the proof of Theorem 3.

D.4 Proof of Lemma 18

Proof. We show that

$$\|P_\omega - P_{\omega'}\|_{\text{TV}} \asymp \zeta r^{\alpha+d+\beta+1}.$$

For ω and ω' that differ only on index j . Recall from Definition 15 that the training data is generated from the joint distribution t_ω , whose density is factored as

$$p(\mathbf{x}) \wp(t \mid \mathbf{x}) \vartheta_{f_\omega}(y; \mathbf{x}, t).$$

Because the context density p and the offline logging propensity $\wp$ do not depend on the packing index ω , the joint distributions differ exclusively in the conditional distribution ϑ_{f_ω} of the outcome Y , which is a Bernoulli random variable with parameter $f_\omega(\mathbf{x}, t)$. Therefore, by the density integral definition of the total variation distance, we have:

$$\|P_\omega - P_{\omega'}\|_{\text{TV}} \tag{45}$$

$$\begin{aligned} &= \frac{1}{2} \int_{\mathcal{X} \times \mathcal{T} \times \mathcal{Y}} |p(\mathbf{x}) \wp(t \mid \mathbf{x}) \vartheta_{f_\omega}(y; \mathbf{x}, t) - p(\mathbf{x}) \wp(t \mid \mathbf{x}) \vartheta_{f_{\omega'}}(y; \mathbf{x}, t)| d\mathbf{x} dt dy \\ &= \int_{\mathcal{X} \times \mathcal{T}} \text{TV}(\text{Ber}(f_\omega(\mathbf{x}, t)), \text{Ber}(f_{\omega'}(\mathbf{x}, t))) \wp(t \mid \mathbf{x}) p(\mathbf{x}) dt d\mathbf{x} \\ &= \int_{\mathcal{X}} \int_{\mathcal{T}} |f_\omega(\mathbf{x}, t) - f_{\omega'}(\mathbf{x}, t)| \wp(t \mid \mathbf{x}) p(\mathbf{x}) dt d\mathbf{x}, \end{aligned} \tag{46}$$

where final equality follows from the fact that the total variation distance between two Bernoulli distributions is exactly the absolute difference between their success probabilities.

Next, we bound this integral using the properties of our distribution instances. The techniques used in this computation is reused in later computations.

$$\begin{aligned} &\int_{\mathcal{X}} \int_{\mathcal{T}} |f_\omega(\mathbf{x}, t) - f_{\omega'}(\mathbf{x}, t)| \wp(t \mid \mathbf{x}) p(\mathbf{x}) dt d\mathbf{x} \\ &\stackrel{(i)}{=} \int_{\bigcup_{j=1}^m B_{r/2}(\mathbf{x}_j^*)} \int_{\tilde{t}-r/2}^{\tilde{t}+r/2} |f_\omega(\mathbf{x}, t) - f_{\omega'}(\mathbf{x}, t)| \wp(t \mid \mathbf{x}) p(\mathbf{x}) dt d\mathbf{x} \\ &\stackrel{(ii)}{=} \int_{B_{r/2}(\mathbf{x}_j^*)} \int_{\tilde{t}-r/2}^{\tilde{t}+r/2} |f_\omega(\mathbf{x}, t) - f_{\omega'}(\mathbf{x}, t)| \wp(t \mid \mathbf{x}) p(\mathbf{x}) dt d\mathbf{x} \\ &\stackrel{(iii)}{\lesssim} r^\beta \int_{B_{r/2}(\mathbf{x}_j^*)} \int_{\tilde{t}-r/2}^{\tilde{t}+r/2} \wp(t \mid \mathbf{x}) p(\mathbf{x}) dt d\mathbf{x} \\ &\stackrel{(iv)}{=} r^\beta \zeta r \int_{B_{r/2}(\mathbf{x}_j^*)} p(\mathbf{x}) d\mathbf{x} \\ &\stackrel{(vi)}{\lesssim} r^\beta \zeta r r^{\alpha+d} = \zeta r^{\alpha+\beta+d+1} \end{aligned}$$

where each step uses:

- (i) Lemma 9,
- (ii) The $r/2$ -balls in $\mathcal{X}$ are disjoint. Additionally, f_ω and $f_{\omega'}$ differ only on the balls corresponding to packing index elements where $\omega_j \neq \omega'_j$.
- (iii) On $B_{r/2}(\mathbf{x}_j^*) \times [\tilde{t} - r/2, \tilde{t} + r/2]$, the difference $|f_\omega - f_{\omega'}|$ is bounded by $2C_\beta r^\beta$. Specifically, difference is $2C_\beta r^\beta$ on $r/4$ -ball and keeps getting smaller as it reaches $r/2$ -ball, due to Lemma 7 and Definition 14. We absorb $2C_\beta$ into the constant.

- (v) The training propensity (Definition 13) is $\wp(t|\mathbf{x}) = \zeta$ on the interval $[\tilde{t} - r/2, \tilde{t} + r/2]$.
- (vi) By the construction of $p(\mathbf{x})$, the density is equal to $C_\alpha r^\alpha$ inside the $r/2$ -balls, and the Lebesgue measure of $r/2$ ball scales as r^d

This completes the verification of $\|P_\omega - P_{\omega'}\|_{\text{TV}} \lesssim \zeta r^{\alpha+d+\beta+1}$. $\square$

E Proof of Non-Private Upper Bound

E.1 Proof of Proposition 1

In the definition of the pessimistic estimator $\tilde{f}_h(\mathbf{x}^Q, p)$, recalled below:

$$\tilde{f}_h(\mathbf{x}^Q, p) := \left(\frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p)}{N_h(\mathbf{x}^Q, p)} - \sqrt{\frac{2 \log 2n}{N_h(\mathbf{x}^Q, p)}} - C_H h^\beta \right) \mathbb{1}(N_h(\mathbf{x}^Q, p) > 0), \quad (47)$$

let us denote the first term as

$$\hat{f}_h(\mathbf{x}, p) := \frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p)}{N_h(\mathbf{x}^Q, p)} \mathbb{1}(N_h(\mathbf{x}^Q, p) > 0) \quad (48)$$

This quantity can be interpreted as an empirical outcome function based on the Nadaraya–Watson (NW) estimator. By construction, $\hat{f}_h(\mathbf{x}, p) = 0$ whenever no training data points lie in the specified neighborhood. In the next lemma, we provide a high-probability pointwise error bound for $\hat{f}_h(\mathbf{x}, p)$.

Proposition 2 (Pointwise error bound). *Suppose Assumption 1. Fix the bandwidth $h > 0$. Then, for any given target context $\mathbf{x}^Q \in \mathcal{X}$ and price candidate $t \in \mathcal{T}$, with probability at least $1 - 1/n$, the following bound holds:*

$$|\hat{f}_h(\mathbf{x}^Q, p) - f(\mathbf{x}^Q, p)| \leq \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\} + \mathbb{1}\{N_h(\mathbf{x}^Q, p) > 0\} \left(Lh^\beta + \sqrt{\frac{2 \log 2n}{N_h(\mathbf{x}^Q, p)}} \right).$$

The proof of Proposition 2 is provided in Appendix E.2.

Proof. We distinguish between the two cases $N_h(\mathbf{x}, p) = 0$ and $N_h(\mathbf{x}, p) > 0$.

Case 1: $N_h(\mathbf{x}, p) = 0$. In this case, the pessimistic estimator evaluates to $\hat{f}_h(\mathbf{x}^Q, p) = 0$. Since the outcome function defined in (??) takes values in $\mathcal{Y} = [0, 1]$, it follows immediately that $\hat{f}_h(\mathbf{x}^Q, p) \leq f(\mathbf{x}^Q, p)$.

Case 2: $N_h(\mathbf{x}, p) > 0$. In this case, the pessimistic estimator evaluates to

$$\tilde{f}_h(\mathbf{x}^Q, p) = \hat{f}_h(\mathbf{x}^Q, p) - \sqrt{\frac{\log n}{N_h(\mathbf{x}^Q, p)}} - h^\beta,$$

and by Proposition 2, with probability at least $1 - 1/n$, the error is bounded as

$$|f(\mathbf{x}^Q, p) - \hat{f}_h(\mathbf{x}^Q, p)| \leq h^\beta + \sqrt{\frac{\log n}{N_h(\mathbf{x}^Q, p)}}.$$

Therefore, we have

$$f(\mathbf{x}^Q, p) \geq \hat{f}_h(\mathbf{x}^Q, p) - h^\beta - \sqrt{\frac{\log n}{N_h(\mathbf{x}^Q, p)}} = \tilde{f}_h(\mathbf{x}^Q, p).$$

$\square$

E.2 Proof of Proposition 2

Proof. Given a target context $\mathbf{x}^Q$ and price candidate $t \in \mathcal{T}$, we have

$$\begin{aligned}
& \left| \hat{f}_h(\mathbf{x}^Q, p) - f(\mathbf{x}^Q, p) \right| \\
&= \left| \frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p)}{N_h(\mathbf{x}^Q, p) \vee 1} - f(\mathbf{x}^Q, p) \right| \\
&= \left| 0 \cdot \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\} + \mathbb{1}\{N_h(\mathbf{x}^Q, p) > 0\} \frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p)}{N_h(\mathbf{x}^Q, p)} - f(\mathbf{x}^Q, p) \right| \\
&= \left| \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\} (0 - f(\mathbf{x}^Q, p)) + \mathbb{1}\{N_h(\mathbf{x}^Q, p) > 0\} \left(\frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p)}{N_h(\mathbf{x}^Q, p)} - f(\mathbf{x}^Q, p) \right) \right| \\
&\leq \underbrace{\left| \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\} (0 - f(\mathbf{x}^Q, p)) \right|}_{E_1} + \underbrace{\left| \mathbb{1}\{N_h(\mathbf{x}^Q, p) > 0\} \left(\frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p)}{N_h(\mathbf{x}^Q, p)} - f(\mathbf{x}^Q, p) \right) \right|}_{E_2},
\end{aligned}$$

where we recall that $I_{h,i}(\mathbf{x}^Q, p) = \mathbb{1}\{(\|\mathbf{X}_i - \mathbf{x}^Q\|_\infty \vee |p_i - p|) \leq h\}$. We bound E_1 and E_2 in order.

Bounding E_1 . Since $\|f\|_\infty \leq 1$, we have

$$E_1 = \left| \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\} (0 - f(\mathbf{x}^Q, p)) \right| = \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\} |f(\mathbf{x}^Q, p)| \leq \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\}.$$

Bounding E_2 . For notational simplicity, from now on we omit $\mathbb{1}\{N_h(\mathbf{x}^Q, p) > 0\}$ multiplied in every term. Using the triangle inequality, we decompose E_2 into two terms:

$$E_2 \leq \underbrace{\left| f(\mathbf{x}^Q, p) - \bar{f} \right|}_{E_{21}} + \underbrace{\left| \bar{f} - \frac{\sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p)}{N_h(\mathbf{x}^Q, p)} \right|}_{E_{22}},$$

where $\bar{f}$ is defined as

$$\bar{f}(\mathbf{x}^Q, p) = \frac{1}{N_h(\mathbf{x}^Q, p)} \sum_{i=1}^n f(\mathbf{x}_i, t_i) I_{h,i}(\mathbf{x}^Q, p).$$

Using the following algebraic equality:

$$f(\mathbf{x}^Q, p) = \frac{1}{N_h(\mathbf{x}^Q, p)} \sum_{i=1}^n f(\mathbf{x}^Q, p) I_{h,i}(\mathbf{x}^Q, p), \quad (49)$$

we can bound E_{21} as follows:

$$\begin{aligned}
E_{21} &= \left| f(\mathbf{x}^Q, p) - \bar{f}(\mathbf{x}^Q, p) \right| \\
&\stackrel{(i)}{\leq} \frac{1}{N_h(\mathbf{x}^Q, p)} \sum_{i=1}^n \left| f(\mathbf{x}^Q, p) - f(\mathbf{X}_i, t_i) \right| I_{h,i}(\mathbf{x}^Q, p) \\
&\stackrel{(ii)}{\leq} \frac{1}{N_h(\mathbf{x}^Q, p)} \sum_{i=1}^n L \left(\|\mathbf{x}_i - \mathbf{x}^Q\|_\infty \vee |t_i - p| \right)^\beta I_{h,i}(\mathbf{x}^Q, p) \\
&\stackrel{(iii)}{\leq} \frac{1}{N_h(\mathbf{x}^Q, p)} \sum_{i=1}^n L h^\beta I_{h,i}(\mathbf{x}^Q, p) \\
&= \frac{1}{N_h(\mathbf{x}^Q, p)} L h^\beta N_h(\mathbf{x}^Q, p) = L h^\beta,
\end{aligned}$$

where step (i) uses (49), step (ii) applies assumptions on conditional outcome in Assumption 1, and step (iii) uses the definition $I_{h,i}(\mathbf{x}^Q, p) = \mathbb{1}\{(\|\mathbf{X}_i - \mathbf{x}^Q\|_\infty \vee |p_i - p|) \leq h\}$.

Next, we bound E_{22} . Using the following algebraic equality:

$$Y_i = \frac{1}{N_h(\mathbf{x}^Q, p)} \sum_{i=1}^n Y_i I_{h,i}(\mathbf{x}^Q, p) \leq h\},$$

we have

$$E_{22} = \left| \frac{1}{N_h(\mathbf{x}^Q, p)} \sum_{i=1}^n (Y_i - f(\mathbf{X}_i, p_i)) I_{h,i}(\mathbf{x}^Q, p) \right|.$$

By the problem setup given in Section 1, conditional on $\{(\mathbf{x}_i, t_i)\}_{i=1}^n$, $(Y_i - f(\mathbf{x}_i, t_i))$'s are zero-mean independent 1-sub-Gaussians, and thus E_2 is the absolute value of the average of $N_h(\mathbf{x}^Q, p)$ zero-mean independent 1-sub-Gaussians. By the sub-Gaussian concentration, with probability at least $1 - 1/n$, we have

$$E_{22} \leq \sqrt{\frac{2 \log 2n}{N_h(\mathbf{x}^Q, p)}}.$$

Conclusion. Collecting the bounds for E_1 , E_{21} and E_{22} , with probability at least $1 - 1/n$, we have

$$|\hat{f}_h(\mathbf{x}^Q, p) - f(\mathbf{x}^Q, p)| \leq \mathbb{1}\{N_h(\mathbf{x}^Q, p) = 0\} + \mathbb{1}\{N_h(\mathbf{x}^Q, p) > 0\} \left(Lh^\beta + \sqrt{\frac{2 \log 2n}{N_h(\mathbf{x}^Q, p)}} \right).$$

This completes the proof of Proposition 2. $\square$

E.3 Proof of Theorem 2

Proof. For a given target context $\mathbf{x}^Q \in \mathcal{X}$, recall that we denote the single optimal price that satisfies Definition 2 as $t^\dagger(\mathbf{x})$. Conditioned on the event from Proposition 2, the suboptimality of the NW-LCB policy relative to any optimal price $t^*(\mathbf{x})$ is bounded by a constant multiple of the lower confidence interval's length. This is shown by the following series of inequalities, that holds with probability at least $1 - 1/n$:

$$\begin{aligned} f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}(\mathbf{x}^Q)) &\stackrel{(i)}{\leq} f(\mathbf{x}, t^*(\mathbf{x})) - \tilde{f}(\mathbf{x}, \hat{\pi}(\mathbf{x})) \\ &\stackrel{(ii)}{=} f(\mathbf{x}, t^\dagger(\mathbf{x})) - \tilde{f}(\mathbf{x}, \hat{\pi}(\mathbf{x})) \\ &\stackrel{(iii)}{\leq} f(\mathbf{x}, t^\dagger(\mathbf{x})) - \tilde{f}(\mathbf{x}, t^\dagger(\mathbf{x})) \\ &\stackrel{(iv)}{\lesssim} h^\beta \mathbb{1}\{N_h(\mathbf{x}, t^\dagger(\mathbf{x})) > 0\} + \log n \frac{\mathbb{1}\{N_h(\mathbf{x}, t^\dagger(\mathbf{x})) > 0\}}{\sqrt{N_h(\mathbf{x}, t^\dagger(\mathbf{x}))}}, \end{aligned}$$

where step (i) Lemma 1 which holds with high probability, step (ii) uses the fact that both of $t^*(\mathbf{x})$ and $t^\dagger(\mathbf{x})$ belong to $\arg \max_{t \in \mathcal{T}} f(\mathbf{x}^Q, p^Q)$, step (iii) uses the definition of NW-LCB policy as the maximizer of $\tilde{f}(\mathbf{x}, p)$ for a fixed $\mathbf{x}$, presented in (6), and step (iv) uses Proposition 2 at point $(\mathbf{x}, t^\dagger(\mathbf{x}))$.

To control the suboptimality, we must control the expected length of the lower confidence interval, taking into account the covariate shift:

$$\begin{aligned}
\text{SubOpt}(\hat{\pi}; Q) &= \\
&\mathbb{E} \left[f(\mathbf{X}^Q, t^\dagger(\mathbf{X}^Q)) - f(\mathbf{X}^Q, \hat{\pi}(\mathbf{X}^Q)) \right], \\
&\lesssim h^\beta \mathbb{E} \left[\mathbb{1} \{ N_h(\mathbf{X}^Q, t^\dagger(\mathbf{X}^Q)) > 0 \} \right] + \log n \mathbb{E} \left[\frac{\mathbb{1} \{ N_h(\mathbf{X}^Q, t^\dagger(\mathbf{X}^Q)) > 0 \}}{\sqrt{N_h(\mathbf{X}^Q, t^\dagger(\mathbf{X}^Q))}} \right] \\
&\leq h^\beta + \log n \mathbb{E} \left[\frac{\mathbb{1} \{ N_h(\mathbf{X}^Q, t^\dagger(\mathbf{X}^Q)) > 0 \}}{\sqrt{N_h(\mathbf{X}^Q, t^\dagger(\mathbf{X}^Q))}} \right]
\end{aligned} \tag{50}$$

We bound the expectation in the last line as follows:

$$\begin{aligned}
\mathbb{E} \left[\frac{\mathbb{1} \{ N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q)) > 0 \}}{\sqrt{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q))}} \right] &\stackrel{(i)}{=} \mathbb{E} \left[\sqrt{\frac{\mathbb{1} \{ N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q)) > 0 \}}{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q))}} \right] \\
&\stackrel{(ii)}{\leq} \sqrt{\mathbb{E} \left[\frac{\mathbb{1} \{ N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q)) > 0 \}}{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q))} \right]}
\end{aligned}$$

where each step uses

(i) the indicator is the same as its square root,

(ii) Jensen's inequality.

The following lemma bounds the expectation in the last display.

Lemma 19. *We have*

$$\mathbb{E} \left[\frac{\mathbb{1} \{ N_h(\mathbf{X}^Q, p^\dagger(\mathbf{X}^Q)) > 0 \}}{N_h(\mathbf{X}^Q, p^\dagger(\mathbf{X}^Q))} \right] \lesssim (\zeta n h)^{-1} h^{-(d+\alpha)}.$$

Proof of Lemma 19 is provided in Appendix E.4.1. In Lemma 19, the expectation is taken with respect to two trainings of randomness the training context-price pair distribution μ , which is used to construct the prediction interval, and the target context distribution Q , which governs the new contexts for which we make a prediction.

Combining (50), (??), and Lemma 19, we have:

$$\text{SubOpt}(\hat{\pi}; Q) \lesssim h^\beta + (n \zeta h)^{-1/2} h^{-(d+\alpha)/2}$$

hiding the log factor. If we set

$$h = \Theta \left((\zeta n)^{-\frac{1}{2\beta+\alpha+d+1}} \right),$$

and

$$\bar{h} = \Omega \left((\zeta n)^{-\frac{1}{2\beta+\alpha+d+1}} \right),$$

we have

$$\begin{aligned}
\text{SubOpt}(\hat{\pi}; Q) &\lesssim (\zeta n)^{-\frac{\beta}{2\beta+\alpha+d+1}} + (\zeta n)^{-1/2} (\zeta n)^{\frac{d+1+\alpha}{2} \frac{1}{2\beta+\alpha+d+1}} \\
&= (\zeta n)^{-\frac{\beta}{2\beta+\alpha+d+1}} + (\zeta n)^{-\frac{1}{2}(1-\frac{d+1+\alpha}{2\beta+\alpha+d+1})} \\
&\lesssim (\zeta n)^{-\frac{\beta}{2\beta+\alpha+d+1}}.
\end{aligned}$$

This completes the proof of Theorem 1. $\square$

E.4 Proof of Supporting Lemmas

E.4.1 Proof of Lemma 19

Proof. To bound the expectations, we apply the law of iterated expectations. Specifically, we first condition on the target context $\mathbf{X}^Q$ and then analyze the inner expectation with respect to the training context-price pairs $(\mathbf{X}_i, t)_{i=1}^n$, which are drawn i.i.d. according to μ . Conditioned on $\mathbf{X}^Q$, the optimal price $t^\dagger(\mathbf{X}^Q)$ becomes deterministic. In this case, the quantity $N_h(\mathbf{X}^Q, t^\dagger(\mathbf{X}^Q))$, which is a transformation of $(\mathbf{X}_i, t)_{i=1}^n$, follows a binomial distribution with number of trials n and success probability

$$\sigma(\mathbf{X}^Q) := \mu(B_h(\mathbf{X}^Q) \times [t^\dagger(\mathbf{X}^Q) - h, t^\dagger(\mathbf{X}^Q) + h]).$$

By Assumption 2, we have the lower bound

$$\sigma(\mathbf{X}^Q) \geq 2\zeta h P(B_h(\mathbf{X}^Q)). \quad (51)$$

Using this distribution and bound, we now proceed to analyze the expectation. apply

$$\begin{aligned}
\mathbb{E} \left[\frac{\mathbb{1}\{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q)) > 0\}}{\sqrt{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q))}} \right] &\stackrel{(i)}{=} \mathbb{E}_Q \left[\mathbb{E}_\mu \left[\frac{\mathbb{1}\{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q)) > 0\}}{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q))} \mid \mathbf{X}^Q \right] \right] \\
&\stackrel{(ii)}{\lesssim} \mathbb{E}_Q \left[\frac{1}{n\sigma(\mathbf{X}^Q)} \right] \\
&\stackrel{(iii)}{\leq} \mathbb{E}_Q \left[\frac{1}{2n\zeta h P(B_h(\mathbf{X}^Q))} \right] \\
&= \frac{1}{2n\zeta h} \mathbb{E}_Q \left[\frac{1}{P(B_h(\mathbf{X}^Q))} \right]. \quad (52)
\end{aligned}$$

where each step uses:

- (i) The law of iterated expectation,
- (ii) Lemma 2, which bounds the ratio of a binomial indicator to the binomial by the inverse expectation,
- (iii) the bound in (51).

All that remains is to control the expectation

$$\mathbb{E}_Q \left[\frac{1}{P(B_h(\mathbf{X}^Q))} \right].$$

To do this, we invoke the transfer exponent assumption (Assumption 1). Let $\mathcal{N}$ denote a minimal $h/2$ -net of $[0, 1]^d$. Using the fact that $|\mathcal{N}| \lesssim h^{-d}$, we have

$$\begin{aligned}
\mathbb{E}_Q \left[\frac{1}{P(B_h(\mathbf{X}^Q))} \right] &= \int_{[0,1]^d} \frac{1}{P(B_h(\mathbf{x}))} dQ(\mathbf{x}) \\
&\stackrel{(i)}{\lesssim} \int_{[0,1]^d} \frac{h^{-\alpha}}{Q(B_h(\mathbf{x}))} dQ(\mathbf{x}) \\
&\stackrel{(ii)}{\leq} h^{-\alpha} \int_{\mathcal{N}} \frac{1}{Q(B_h(\mathbf{x}))} dQ(\mathbf{x}) \\
&= h^{-\alpha} \sum_{\mathbf{z} \in \mathcal{N}} \int_{B_{h/2}(\mathbf{z})} \frac{1}{Q(B_h(\mathbf{x}))} dQ(\mathbf{x}) \\
&\stackrel{(iii)}{\leq} h^{-\alpha} \sum_{\mathbf{z} \in \mathcal{N}} \int_{B_{h/2}(\mathbf{z})} \frac{1}{Q(B_{h/2}(\mathbf{z}))} dQ(\mathbf{x}) \\
&= h^{-\alpha} \sum_{\mathbf{z} \in \mathcal{N}} \frac{1}{Q(B_{h/2}(\mathbf{z}))} \int_{B_{h/2}(\mathbf{z})} dQ(\mathbf{x}) \\
&= h^{-\alpha} \sum_{\mathbf{z} \in \mathcal{N}} \frac{1}{Q(B_{h/2}(\mathbf{z}))} Q(B_{h/2}(\mathbf{z})) \\
&= h^{-\alpha} \cdot |\mathcal{N}| \\
&\lesssim h^{-(d+\alpha)},
\end{aligned} \tag{53}$$

where each step uses:

- (i) $\kappa(P, Q) \leq \alpha$,
- (ii) $[0, 1]^d \subset \bigcup_{\mathbf{z} \in \mathcal{N}} B_{h/2}(\mathbf{z})$
- (iii) $\mathbf{x} \in B_{h/2}(\mathbf{z})$ implies $B_{h/2}(\mathbf{z}) \subset B_h(\mathbf{x})$, thus we have $1/Q(B_h(\mathbf{x})) \leq 1/Q(B_{h/2}(\mathbf{z}))$

Therefore, combining (52) and (53), we have the following bound

$$\mathbb{E} \left[\frac{\mathbb{1}\{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q)) > 0\}}{N_h(\mathbf{X}^Q, p^*(\mathbf{X}^Q))} \right] \lesssim (\zeta n h)^{-1} h^{-(d+\alpha)}.$$

□

F Proof of Private Upper Bound

F.1 Proof of Theorem 4

Proof. For notational simplicity, we fix an arbitrary query point $(\mathbf{x}^Q, t^Q)$ and suppress the dependency on $(\mathbf{x}^Q, t^Q)$ in our notation where the context is clear. Specifically, we define the following notations:

- $j = j(\mathbf{x}^Q, t^Q)$: the index of the bin containing the query point.

- $\mu := \bar{f}_n(\mathbf{x}^Q, t^Q)$: The expected non-private estimator value.
- $N := \sum_{i=1}^n U_{i,j}$: The true (unobserved) sample count of the j -th bin.
- $\tilde{N} := \tilde{N}_{h,j}(\mathbf{x}^Q, t^Q) = \sum_{i=1}^n \tilde{U}_{i,j}$: The observed count of the privatized sample in the j -th bin.
- $\tilde{f}_{n,\varepsilon} = \tilde{f}_{n,\varepsilon}(\mathbf{x}^Q, t^Q)$: the private conditional outcome estimator evaluated at $\mathbf{x}^Q, t^Q$

The estimation error of the private estimator can be expressed as:

$$|\tilde{f}_{n,\varepsilon} - \mu| = \left| \frac{\sum_{i=1}^n \tilde{V}_{i,j}}{\tilde{N}} - \mu \right| = \left| \frac{\sum_{i=1}^n \tilde{V}_{i,j} - \mu \tilde{N}}{\tilde{N}} \right|.$$

Because the Laplace distribution is continuous, $\tilde{N} \neq 0$ almost surely, ensuring that division by $\tilde{N}$ is well-defined. Substituting the definitions $\tilde{U}_{i,j} = U_{i,j} + \zeta_{i,j}$ and $\tilde{V}_{i,j} = V_{i,j} + \xi_{i,j}$, where the privacy noise terms $\zeta_{i,j}, \xi_{i,j} \stackrel{iid}{\sim} \text{Lap}(0, \sigma_\varepsilon^2/2)$, we decompose the numerator into a distinct sampling error component and a privacy error component:

$$\begin{aligned} \sum_{i=1}^n \tilde{V}_{i,j} - \mu \tilde{N} &= \sum_{i=1}^n \tilde{V}_{i,j} - \mu \sum_{i=1}^n \tilde{U}_{i,j} = \sum_{i=1}^n (V_{i,j} + \xi_{i,j}) - \mu \sum_{i=1}^n (U_{i,j} + \zeta_{i,j}) \\ &= \underbrace{\sum_{i=1}^n (V_{i,j} - \mu U_{i,j})}_{:= S_n \text{ sampling error}} + \underbrace{\sum_{i=1}^n (\xi_{i,j} - \mu \zeta_{i,j})}_{:= P_n \text{ privacy error}} \end{aligned}$$

We now establish high-probability bounds for each error component. By bounding S_n and P_n independently, we will demonstrate that

$$\Pr \left(|S_n| \leq \sqrt{\frac{1}{2} N \log n} \right) \geq 1 - \frac{2}{n} \text{ and } \Pr \left(|P_n| \leq 2 \frac{4\sqrt{2}}{\varepsilon} \sqrt{n \log n} \right) \geq 1 - \frac{4}{n}.$$

Bounding the Sampling Error (S_n). We analyze the sampling error by first conditioning on $\mathcal{F}_n := \{\mathbf{X}_i, T_i\}_{i=1}^n$, and then using the law of total expectation. Under this conditioning, the bin indicators $U_{i,j}$ and the bin count $N = \sum_i U_{i,j}$ and μ are deterministic constants. The randomness arises solely from the outcomes Y_i .

Conditioned on $N > 0$, the primary tool employed here is Hoeffding's inequality. We begin by demonstrating that S_n has a mean of zero. We start by expressing S_n as

$$S_n = \sum_{i=1}^n (V_{i,j} - \mu U_{i,j}) = \sum_{i=1}^n U_{i,j} (Y_i - \mu) = \sum_{i=1}^n U_{i,j} Y_i - \mu \sum_{i=1}^n U_{i,j},$$

where, conditioned on $\mathcal{F}_n$, the first term is random and the second term is deterministic. It is clear that S_n is centered, because

$$\mathbb{E} \left[\sum_{i=1}^n U_{i,j} Y_i | \mathcal{F}_n \right] = \sum_{i=1}^n U_{i,j} \mathbb{E}[Y_i | \mathcal{F}_n] = \sum_{i=1}^n U_{i,j} f(\mathbf{X}_i, T_i) = \mu \sum_{i=1}^n U_{i,j}.$$

Next, we show that S_n is a bounded random variable, conditioned on $\mathcal{F}_n$. From the expression $S_n = \sum_{i=1}^n U_{i,j} (Y_i - \mu)$, we can see that S_n is a sum of independent terms

$$Z_i := U_{i,j} (Y_i - \mu).$$

Given that $U_{i,j} \in \{0, 1\}$, the number of non-zero terms is bounded above by N , where $N \leq n$. We define the terms for which $U_{i,j} = 1$ as *active*. Because the variables Y_i are bounded in $[0, 1]$, each active Z_i falls within an interval $[-\mu, 1 - \mu]$ of length 1. Consequently, every active Z_i is $(1/4)$ -sub-Gaussian. Applying Hoeffding's inequality to the sum of these N active terms yields:

$$\Pr(|S_n| \geq \delta \mid \mathcal{F}_n, N > 0) \leq 2 \exp\left(-\frac{\delta^2}{2(N/4)}\right) = 2 \exp\left(-\frac{2\delta^2}{N}\right).$$

We set the threshold $\delta = \sqrt{\frac{1}{2}N \log n}$ to get:

$$\Pr\left(|S_n| \leq \sqrt{\frac{1}{2}N \log n} \mid \mathcal{F}_n, N > 0\right) \geq 1 - 2 \exp\left(-\frac{2(\frac{1}{2}N \log n)}{N}\right) = 1 - \frac{2}{n}.$$

If we condition on $N = 0$, then $S_n = 0$ and $\frac{1}{2}N \log n = 0$, thus the probability is 1, and the inequality above is satisfied. Since the above inequality holds for all value of N , we have

$$\Pr\left(|S_n| \leq \sqrt{\frac{1}{2}N \log n}\right) = \mathbb{E}\left[\Pr\left(|S_n| \leq \sqrt{\frac{1}{2}N \log n} \mid \mathcal{F}_n\right)\right] \geq \mathbb{E}\left[1 - \frac{2}{n}\right] = 1 - \frac{2}{n}.$$

Thus, with probability at least $1 - 2/n$, the sampling error satisfies:

$$|S_n| \leq \sqrt{\frac{1}{2}N \log n}. \quad (54)$$

Bounding the Privacy Error (P_n). Let Z_V and Z_U denote the sums of n i.i.d. Laplace variables with scale $b = \sigma_\varepsilon/\sqrt{2}$. The privacy error P_n satisfies

$$|P_n| \leq |Z_V| + \mu|Z_U| \leq |Z_V| + |Z_U|,$$

where the last inequality uses $\mu \leq 1$.

Applying the sub-exponential tail bounds:

$$\begin{aligned} P(|Z_U| \geq \tau_{n,\varepsilon}) &\leq 2 \exp\left(-\min\left(\frac{\tau_{n,\varepsilon}^2}{4nb^2}, \frac{\tau_{n,\varepsilon}}{2b}\right)\right), \\ P(|Z_V| \geq \tau_{n,\varepsilon}) &\leq 2 \exp\left(-\min\left(\frac{\tau_{n,\varepsilon}^2}{4nb^2}, \frac{\tau_{n,\varepsilon}}{2b}\right)\right) \end{aligned}$$

We set the failure probability to $2/n$, requiring the exponent to be at least $\log n$. If we choose the threshold

$$\tau_{n,\varepsilon} := b\sqrt{4n \log n}, \quad (55)$$

substituting this threshold into the Gaussian term and exponential term of the minimum becomes

$$\frac{\tau_{n,\varepsilon}^2}{4nb^2} = \frac{4nb^2 \log n}{4nb^2} = \log n \quad \frac{\tau_{n,\varepsilon}}{2b} = \frac{b\sqrt{4n \log n}}{2b} = \sqrt{n \log n}.$$

Denote these concentration events as $\mathcal{E}_\xi$ and $\mathcal{E}_\zeta$, respectively. Since $\log n \leq \sqrt{n \log n}$ for all $n \geq 1$, the Gaussian term is smaller and thus determines the probability bound. Substituting $b = \sigma_\varepsilon/\sqrt{2} = 4/\varepsilon$, the threshold becomes:

$$\tau_{n,\varepsilon} = \frac{4\sqrt{2}}{\varepsilon} \sqrt{n \log n}. \quad (56)$$

Thus, with probability at least $1 - 4/n$:

$$|P_n| \leq 2\tau_{n,\varepsilon} \quad \text{and} \quad |N - \tilde{N}| = |Z_U| \leq \tau_{n,\varepsilon}. \quad (57)$$

Combination of the Bounds. By the union bound, with probability at least $1 - 6/n$, the total numerator error is bounded by:

$$\begin{aligned}
|S_n| + |P_n| &\leq \sqrt{\frac{1}{2}N \log n + 2\tau_{n,\varepsilon}} \\
&\leq \sqrt{\frac{1}{2}(\tilde{N} + \tau_{n,\varepsilon}) \log n + 2\tau_{n,\varepsilon}} \\
&\leq \sqrt{\frac{1}{2}|\tilde{N}| \log n} + \sqrt{\frac{1}{2}\tau_{n,\varepsilon} \log n} + 2\tau_{n,\varepsilon},
\end{aligned} \tag{58}$$

where the second inequality uses $|N - \tilde{N}| = |Z_U| \leq \tau_{n,\varepsilon}$. In the regime where $\varepsilon \leq 8\sqrt{2}\sqrt{n/\log n}$, it holds that $\sqrt{\frac{1}{2}\tau_{n,\varepsilon} \log n} \leq \tau_{n,\varepsilon}$. Consequently, we have:

$$|S_n| + |P_n| \leq \sqrt{\frac{1}{2}|\tilde{N}| \log n} + 3\tau_{n,\varepsilon}. \tag{59}$$

Dividing by $|\tilde{N}|$ yields the result:

$$|\tilde{f}_{n,\varepsilon} - \mu| \leq \sqrt{\frac{\log n}{2}} \frac{1}{\sqrt{|\tilde{N}|}} + 3\frac{\tau_{n,\varepsilon}}{|\tilde{N}|}.$$

Substituting the definition of $\tau_{n,\varepsilon}$ completes the proof of Theorem 4. $\square$

F.2 Proof of Lemma 1

Proof. Fix the query point $(\mathbf{x}^Q, t^Q)$. To streamline the presentation, we suppress the explicit dependence on the query point $(\mathbf{x}^Q, t^Q)$, the bandwidth h , and the privacy parameter ε where the context is clear. Specifically, letting j denote the index of the bin containing $(\mathbf{x}^Q, t^Q)$, we adopt the following shorthand notations for the true function value, the empirical non-private average, and the private LDP estimator:

- $j = j(\mathbf{x}^Q, t^Q)$: the index of the bin containing the query point.
- $f := f(\mathbf{x}^Q, t^Q)$
- $\mu := \bar{f}_n(\mathbf{x}^Q, t^Q) = \frac{\sum_{i=1}^n f(\mathbf{X}_i, T_i) U_{i,j}(\mathbf{x}^Q, t^Q)}{\sum_{i=1}^n U_{i,j}(\mathbf{x}^Q, t^Q)}$
- $\tilde{f} := \tilde{f}_{n,\varepsilon}(\mathbf{x}^Q, t^Q) = \frac{\sum_{i=1}^n \tilde{V}_{i,j}(\mathbf{x}^Q, t^Q)}{\sum_{i=1}^n \tilde{U}_{i,j}(\mathbf{x}^Q, t^Q)}$
- $\tilde{N} := \tilde{N}_{h,j(\mathbf{x}^Q, t)} = \sum_{i=1}^n \tilde{U}_{i,j}$: The observed privatized count.
- $W := C_{\text{LDP}} \frac{\sqrt{\max\{n/\varepsilon^2, \tilde{N}\} \log n}}{\tilde{N}}$: confidence width defined in Theorem 4

We condition on the "good event" $\mathcal{E} := \{|\tilde{f} - \mu| \leq W\}$ which is the event where the concentration bound from Theorem 4 holds. Theorem 4 guarantees that $\mathbb{P}(\mathcal{E}) \geq 1 - 6/n$.

If $\tilde{N} < \tau_{n,\varepsilon}$, the indicator function in (15) is zero, and the pessimistic estimator returns 0. Since outcomes are non-negative ($f \geq 0$), the inequality $\tilde{f}_{n,\varepsilon,h} \leq f$ holds trivially.

Now, assume $\tilde{N} \geq \tau_{n,\varepsilon}$. We decompose the deviation from the true conditional outcome value $f(\mathbf{x}^Q, t)$ as:

$$\tilde{f} - f = (\tilde{f} - \mu) + (\mu - f).$$

For the first part, since we condition on $\mathcal{E}$, we can apply the concentration bound:

$$\tilde{f} - \mu \leq |\tilde{f} - \mu| \leq W.$$

To bound the second term $\mu - f$, we first observe that the expected non-private estimator value μ is a convex combination of the true function values evaluated at the sample points within the bin $A_{h,j}$. Let us define the normalized weights

$$w_i = \frac{U_{i,j}}{\sum_{k=1}^n U_{k,j}}.$$

Since $U_{i,j} \geq 0$, it follows that $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$. Consequently, we can express $\mu - f$ as

$$\mu - f = \sum_{i=1}^n w_i f(\mathbf{X}_i, T_i) - f = \sum_{i=1}^n w_i (f(\mathbf{X}_i, T_i) - f).$$

Taking the absolute value and applying the triangle inequality yields

$$|\mu - f| \leq \sum_{i=1}^n w_i |f(\mathbf{X}_i, T_i) - f|.$$

By definition, the weights w_i are exactly zero for any observation $(\mathbf{X}_i, T_i) \notin A_{h,j}$. Thus, the absolute difference for any point contributing to the summation is strictly bounded by the maximum possible variation of f within the bin $A_{h,j}$. Bounding the terms inside the sum by the supremum over B_j , we obtain

$$\begin{aligned} |\mu - f| &\leq \sum_{i=1}^n w_i |f(\mathbf{X}_i, T_i) - f| \\ &\leq \sup_{(\mathbf{x}, t) \in A_{h,j}} \left(\sum_{i=1}^n w_i |f(\mathbf{x}, t) - f| \right) \\ &= \left(\sup_{(\mathbf{x}, t) \in A_{h,j}} |f(\mathbf{x}, t) - f(\mathbf{x}^Q, t^Q)| \right) \sum_{i=1}^n w_i = \sup_{(\mathbf{x}, t) \in A_{h,j}} |f(\mathbf{x}, t) - f(\mathbf{x}^Q, t^Q)|. \end{aligned}$$

Finally, we apply the Hölder continuity condition (Assumption 1). For any point $(\mathbf{x}, t) \in A_{h,j}$, its distance to the query point $(\mathbf{x}^Q, t^Q) \in A_{h,j}$ is bounded by the bin width parameter h . This spatial constraint implies that

$$\sup_{(\mathbf{x}, t) \in A_{h,j}} |f(\mathbf{x}, t) - f(\mathbf{x}^Q, t^Q)| \leq C_H h^\beta.$$

Therefore we have

$$\mu - f \leq |\mu - f| \leq C_H h^\beta.$$

Combining the two bounds, we have, with probability at least $1 - 6/n$:

$$\tilde{f} - f \leq W + C_H h^\beta \iff f \geq \tilde{f} - W - C_H h^\beta,$$

where the right-hand side corresponds to the pessimistic estimator. This completes the proof of Lemma 1. $\square$

F.3 Supporting Lemmas

F.3.1 Proof of Lemma ??

Proof. For notational simplicity, we define the following quantities:

- $N := \sum_{i=1}^n U_{i,j}$: The true (latent) sample count.
- $Z := \sum_{i=1}^n \zeta_{i,j}$: The total privacy noise.
- $\tilde{N} := N + Z$: The observed privatized count.
- $\tau := \tau_{n,\varepsilon} = \frac{4\sqrt{2}}{\varepsilon} \sqrt{n \log n}$.

Throughout the proof, we condition on the "good event" of Theorem 4, denoted $\mathcal{E}$, which guarantees that $\{|Z| \leq \tau\}$.

Step 1: Denominator. Consider the quantity on the left-hand side. The indicator $\mathbb{1}\{\tilde{N} > \tau\}$ implies $\tilde{N} > 0$, so the term is well-defined. We begin by expanding the ratio using the identity $N = \tilde{N} - Z$:

$$\frac{\mathbb{1}\{\tilde{N} > \tau\}}{\tilde{N}} = \frac{\mathbb{1}\{\tilde{N} > \tau\}}{N} \cdot \frac{N}{\tilde{N}} = \frac{\mathbb{1}\{\tilde{N} > \tau\}}{N} \cdot \frac{\tilde{N} - Z}{\tilde{N}}.$$

Applying the triangle inequality $|\tilde{N} - Z| \leq |\tilde{N}| + |Z|$ (and noting that $\tilde{N} > 0$ under the indicator):

$$\frac{\mathbb{1}\{\tilde{N} > \tau\}}{N} \cdot \frac{|\tilde{N} - Z|}{\tilde{N}} \leq \frac{\mathbb{1}\{\tilde{N} > \tau\}}{N} \left(\frac{\tilde{N} + |Z|}{\tilde{N}} \right) = \frac{\mathbb{1}\{\tilde{N} > \tau\}}{N} \left(1 + \frac{|Z|}{\tilde{N}} \right).$$

Using the event $\mathcal{E}$ ($|Z| \leq \tau$) and the domain of the indicator ($\tilde{N} > \tau$), the ratio $|Z|/\tilde{N}$ is strictly bounded by 1:

$$\frac{|Z|}{\tilde{N}} < \frac{\tau}{\tau} = 1.$$

Substituting this back yields:

$$\frac{\mathbb{1}\{\tilde{N} > \tau\}}{\tilde{N}} \leq \frac{2}{N} \mathbb{1}\{\tilde{N} > \tau\}.$$

Numerator. Conditioned on $\mathcal{E}$, if the observed count exceeds the noise threshold ($\tilde{N} > \tau$), then the true count must be positive. Specifically:

$$\tilde{N} > \tau \iff N + Z > \tau \iff N > \tau - Z.$$

Since $|Z| \leq \tau$, we have $Z \leq \tau$, and thus $\tau - Z \geq 0$. More strictly, since N is an integer, if $N > 0$ were false (i.e., $N = 0$), then we would require $Z > \tau$, which contradicts event $\mathcal{E}$. Thus, on event $\mathcal{E}$:

$$\mathbb{1}\{\tilde{N} > \tau\} \leq \mathbb{1}\{N > 0\}.$$

Conclusion. Combining Step 1 and Step 2, we obtain the final bound:

$$\frac{\mathbb{1}\{\tilde{N} > \tau\}}{\tilde{N}} \leq \frac{2}{N} \mathbb{1}\{\tilde{N} > \tau\} \leq \frac{2}{N} \mathbb{1}\{N > 0\}.$$

□

F.4 aaa

Given target context $\mathbf{x}^Q \in \mathcal{X}$, recall from Assumption ?? that $t^\dagger(\mathbf{x}^Q)$ denotes the single optimal price that satisfies the near-optimal price exploration. Let j be the bin index of the bin containing the pair $(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q))$. Let us denote the non-private empirical measures for the bin containing $(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q))$ as

$$\hat{\mu} := \mu_n(A_{h,j}), \quad \nu := \nu_n(A_{h,j}),$$

and the private (noisy) versions as

$$\tilde{\mu} = \tilde{\mu}_n(A_{h,j}), \quad \tilde{\nu} = \tilde{\nu}_n(A_{h,j}).$$

Denote the small concentration constant

$$A := \frac{2 \log(2n)}{n}.$$

and recall that

$$t_\varepsilon = \frac{16}{\varepsilon} \sqrt{\frac{\log n}{n}}.$$

The analysis begins by splitting $f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q))$ into two cases with respect to $\hat{\mu}$:

$$\begin{aligned} f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q)) &= \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \{f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q))\} \\ &\quad + \mathbb{1}\{\hat{\mu} < 4t_\varepsilon\} \{f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q))\} \\ &\lesssim \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \{f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q))\} + \mathbb{1}\{\hat{\mu} < 4t_\varepsilon\}. \end{aligned}$$

We first bound the expectation of the second term.

Lemma 20. *If we set h as $h = \Omega((\varepsilon^2 \zeta^2 n)^{-\frac{1}{2(\alpha+d+1)}})$ Then we have*

$$\mathbb{E}[\mathbb{1}\{\hat{\mu} < 4t_\varepsilon\}] \lesssim (n \zeta h)^{-1} h^{-(d+\alpha)}.$$

Proof of Lemma 20 is provided in Appendix F.6.

Conditioned on the event in Proposition ?? (which occurs with probability at least $1 - 4/n$), the suboptimality of the policy at $(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q))$, relative to any optimal price $t^*(\mathbf{x}^Q)$, can be decomposed into estimation and privacy errors through the following sequence of inequalities:

$$\begin{aligned} &f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q)) \\ &\stackrel{(i)}{\leq} f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - \tilde{f}_{h,\text{DP}}(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q)) \\ &\stackrel{(ii)}{=} f(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) - \tilde{f}_{h,\text{DP}}(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q)) \\ &\stackrel{(iii)}{\leq} f(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) - \tilde{f}_{h,\text{DP}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) \\ &= \{f(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) - \tilde{f}_{h,\text{BIN}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q))\} + \{\tilde{f}_{h,\text{BIN}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) - \tilde{f}_{h,\text{DP}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q))\} \end{aligned} \tag{60}$$

where each step uses

- (i) Proposition ??, conditioned on the noise concentration event which occurs with probability at least $1 - 4/n$,
- (ii) Both of $t^*(\mathbf{x})$ and $t^\dagger(\mathbf{x})$ belong to the set $\arg \max_{t \in \mathcal{T}} f(\mathbf{x}^Q, p^Q)$,
- (iii) The definition of the policy as the maximizer of $\tilde{f}_{h,\text{DP}}(\mathbf{x}, p)$ for a fixed $\mathbf{x}$, presented in (??).

By the same argument as in Appendix ??, we have

$$\mathbb{E}[f(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) - \tilde{f}_{h,\text{BIN}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q))] \lesssim h^\beta + (n \zeta h)^{-1/2} h^{-(d+\alpha)/2} \quad (61)$$

is bounded. Therefore we focus on bounding the expectation of

$$\Delta := \tilde{f}_{h,\text{BIN}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) - \tilde{f}_{h,\text{DP}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)),$$

where we recall from (47) that

$$\tilde{f}_{h,\text{BIN}}(\mathbf{x}^Q, p) = \left(\frac{\nu_n(A_{h,j})}{\mu_n(A_{h,j})} - \sqrt{\frac{(2 \log 2n)/n}{\mu_n(A_{h,j})}} - Lh^\beta \right) \mathbb{1}(\mu_n(A_{h,j}) > 0).$$

For simplicity, we adapt the following notations. First, we omit the dependence on $(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q))$, since all functions are evaluated at this point.

Using these notations, we have

$$\tilde{f}_{h,\text{DP}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) = \left(\frac{\tilde{\nu} - t_\varepsilon}{\tilde{\mu} + t_\varepsilon} - \sqrt{\frac{A}{\tilde{\mu} - t_\varepsilon}} - Lh^\beta \right) \mathbb{1}(\tilde{\mu} > t_\varepsilon),$$

and

$$\tilde{f}_{h,\text{BIN}}(\mathbf{x}^Q, t^\dagger(\mathbf{x}^Q)) = \left(\frac{\nu}{\hat{\mu}} - \sqrt{\frac{A}{\hat{\mu}}} - Lh^\beta \right) \mathbb{1}(\hat{\mu} > 0).$$

Thus we can write Δ as

$$\Delta = \left(\frac{\nu}{\hat{\mu}} - \sqrt{\frac{A}{\hat{\mu}}} - Lh^\beta \right) \mathbb{1}(\hat{\mu} > 0) - \left(\frac{\tilde{\nu} - t_\varepsilon}{\tilde{\mu} + t_\varepsilon} - \sqrt{\frac{A}{\tilde{\mu} - t_\varepsilon}} - Lh^\beta \right) \mathbb{1}(\tilde{\mu} > t_\varepsilon).$$

Lemma 21. *With probability at least $1 - 4/n$, we have*

$$\mathbb{E}[\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \Delta] \lesssim (t_\varepsilon \vee \sqrt{At_\varepsilon})(\zeta n h)^{-1} h^{-(d+\alpha)}$$

Proof of Lemma 21 is provided in Appendix F.5.

Collecting the bounds (61), Lemma 21, and Lemma 20, we have, for $h = \Omega((\varepsilon^2 \zeta^2 n)^{-\frac{1}{2(\alpha+d+1)}})$,

$$\mathbb{E}[f(\mathbf{x}^Q, t^*(\mathbf{x}^Q)) - f(\mathbf{x}^Q, \hat{\pi}_{h,\text{DP}}(\mathbf{x}^Q))] \lesssim h^\beta + (n \zeta h)^{-1/2} h^{-(d+\alpha)/2} + (t_\varepsilon \vee \sqrt{At_\varepsilon})(\zeta n h)^{-1} h^{-(d+\alpha)}$$

(T_1) **Bias Term:** This term arises from the Hölder smoothness of f and scales as:

$$T_1 \approx h^\beta$$

(T_2) **Estimation/Exploration Variance Term:** This term represents the non-private statistical error under covariate shift and imperfect exploration (ζ). It simplifies to:

$$T_2 \approx (\zeta n)^{-1/2} h^{-\frac{d+\alpha+1}{2}}$$

(T_3) **Privacy/Noise Term:** This term captures the additional variance introduced by the LDP mechanism. Using the rates $t_\varepsilon \approx \tilde{O}(\varepsilon^{-1} n^{-1/2})$ and $A \approx \tilde{O}(n^{-1})$, the dominant term is t_ε , leading to the dependence on n as $n^{-3/2}$.

$$T_3 \approx (\varepsilon^{-1} n^{-1/2})(\zeta n)^{-1} h^{-(d+\alpha+1)} = (\varepsilon \zeta n^{3/2})^{-1} h^{-(d+\alpha+1)}$$

Balance Point h_{12} (Bias vs. Estimation) The first candidate bandwidth h_{12} is found by balancing the bias term T_1 and the estimation term T_2 :

$$T_1 \stackrel{!}{=} T_2 \implies h^\beta = (\zeta n)^{-1/2} h^{-\frac{d+\alpha+1}{2}}$$

Solving for h :

$$h^{\beta+\frac{d+\alpha+1}{2}} = (\zeta n)^{-1/2} \implies h_{12} = (\zeta n)^{-\frac{1}{2\beta+d+\alpha+1}}$$

The corresponding rate is:

$$\text{Rate}_{12} \approx (\zeta n)^{-\frac{\beta}{2\beta+d+\alpha+1}}$$

Balance Point h_{13} (Bias vs. Privacy/Noise) The second candidate bandwidth h_{13} is found by balancing the bias term T_1 and the privacy/noise term T_3 :

$$T_1 \stackrel{!}{=} T_3 \implies h^\beta = (\varepsilon \zeta n^{3/2})^{-1} h^{-(d+\alpha+1)}$$

Solving for h :

$$h^{\beta+d+\alpha+1} = (\varepsilon \zeta n^{3/2})^{-1} \implies h_{13} = (\varepsilon \zeta n^{3/2})^{-\frac{1}{\beta+d+\alpha+1}}$$

The corresponding rate is:

$$\text{Rate}_{13} \approx (\varepsilon \zeta n^{3/2})^{-\frac{\beta}{\beta+d+\alpha+1}}$$

Thus choosing the bandwidth $h_{12} \vee h_{13}$ yields the rate

$$\approx (\zeta n)^{-\frac{\beta}{2\beta+d+\alpha+1}} \vee (\varepsilon \zeta n^{3/2})^{-\frac{\beta}{\beta+d+\alpha+1}}$$

The privacy term dominates the overall sub-optimal gap when the privacy budget ε is smaller than the following critical threshold.

$$\varepsilon \lesssim \zeta^{-\frac{\beta}{2\beta+d+\alpha+1}} \cdot n^{-\frac{4\beta+d+\alpha+1}{4\beta+2d+2\alpha+2}}$$

F.5 Proof of Lemma 21

Proof. Let us denote the noise concentration event as

$$E_\varepsilon := \left\{ |\text{AN}_\nu| \leq t_\varepsilon, |\text{AN}_\mu| \leq t_\varepsilon \right\}.$$

If $\hat{\mu} > 4t_\varepsilon$, we have

$$\tilde{\mu} = \hat{\mu} + \text{AN}_\mu > \hat{\mu} - t_\varepsilon > 3t_\varepsilon > t_\varepsilon$$

Therefore, all indicators in Δ are gone and we have

$$\Delta = \underbrace{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \left(\frac{\nu}{\hat{\mu}} - \frac{\tilde{\nu} - t_\varepsilon}{\tilde{\mu} + t_\varepsilon} \right)}_{\Delta_1} + \underbrace{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \sqrt{A} \left(\frac{1}{\sqrt{\tilde{\mu}} - t_\varepsilon} - \frac{1}{\sqrt{\hat{\mu}}} \right)}_{\Delta_2}.$$

Bounding $\mathbb{E}[\Delta_1]$. Note that for $b, d \neq 0$,

$$\begin{aligned} \frac{a}{b} - \frac{c}{d} &= \frac{ad - bc}{bd} = \frac{ad - cd + cd - bc}{bd} = \frac{(a - c)d + c(d - b)}{bd} = \frac{a - c}{b} \frac{d}{d} + \frac{c}{bd} (d - b) \\ &= \frac{a - c}{b} + \frac{c}{bd} (d - b). \end{aligned}$$

Hence, conditioned on E_ε , we have

$$\begin{aligned}
\Delta_1 &= \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \left\{ \frac{\hat{\nu} - \tilde{\nu} + t_\varepsilon}{\hat{\mu}} + \frac{\tilde{\mu} - \hat{\mu} + t_\varepsilon}{\hat{\mu}(\tilde{\mu} + t_\varepsilon)} (\tilde{\nu} - t_\varepsilon) \right\} \\
&= \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \left\{ \frac{-\text{AN}_\nu + t_\varepsilon}{\hat{\mu}} + \frac{\text{AN}_\mu + t_\varepsilon}{\hat{\mu}} \frac{\tilde{\nu} - t_\varepsilon}{\tilde{\mu} + t_\varepsilon} \right\} \\
&= \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \left[\frac{1}{\hat{\mu}} \left\{ (-\text{AN}_\nu + t_\varepsilon) + \frac{\tilde{\nu} - t_\varepsilon}{\tilde{\mu} + t_\varepsilon} (\text{AN}_\mu + t_\varepsilon) \right\} \right] \\
&= \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \left[\frac{1}{\hat{\mu}} \left\{ (-\text{AN}_\nu + t_\varepsilon) + \frac{\hat{\nu} + \text{AN}_\nu - t_\varepsilon}{\hat{\mu} + \text{AN}_\mu + t_\varepsilon} (\text{AN}_\mu + t_\varepsilon) \right\} \right] \\
&\leq \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}} |(-\text{AN}_\nu + t_\varepsilon)| + \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}} \left| \frac{\hat{\nu} + \text{AN}_\nu - t_\varepsilon}{\hat{\mu} + \text{AN}_\mu + t_\varepsilon} \right| |\text{AN}_\mu + t_\varepsilon| \\
&\stackrel{(i)}{\leq} \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}} |(-\text{AN}_\nu + t_\varepsilon)| + \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}} \left| \frac{\hat{\nu} + \text{AN}_\nu - t_\varepsilon}{\hat{\mu}} \right| |\text{AN}_\mu + t_\varepsilon| \\
&\stackrel{(ii)}{\leq} 2t_\varepsilon \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}} + 2t_\varepsilon \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}^2} \left| \frac{\hat{\nu}}{\hat{\mu}} \right| + 4t_\varepsilon^2 \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}^2} \\
&\stackrel{(iii)}{\leq} 4t_\varepsilon \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}} + 4t_\varepsilon^2 \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}^2} \\
&\stackrel{(iv)}{\leq} 5t_\varepsilon \frac{\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}}{\hat{\mu}} \\
&\leq 5t_\varepsilon \frac{\mathbb{1}\{\hat{\mu} > 0\}}{\hat{\mu}}
\end{aligned}$$

where

- (i) $|\hat{\mu} + \text{AN}_\mu + t_\varepsilon| \geq |\hat{\mu} - t_\varepsilon + t_\varepsilon| = |\hat{\mu}|$ under the noise concentration event E_ε ,
- (ii) the noise concentration event E_ε
- (iii) $\hat{\mu} < \hat{\nu}$ because $0 \leq Y_i \leq 1$ and we defined the empirical measures in (??) as
$$\hat{\mu} = \mu_n(A_{h,j}) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}((X_i, p_i) \in A_{h,j}), \quad \hat{\nu} = \nu_n(A_{h,j}) = \frac{1}{n} \sum_{i=1}^n Y_i \mathbb{1}((X_i, p_i) \in A_{h,j}).$$
- (iv) $\mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} (2t_\varepsilon/\hat{\mu}) < \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\}$.

Using the same argument as in the proof of Lemma 19, we have

$$\mathbb{E}[\Delta_1] \lesssim t_\varepsilon (\zeta n h)^{-1} h^{-(d+\alpha)}.$$

Bounding $\mathbb{E}[\Delta_2]$. If $\hat{\mu} > 4t_\varepsilon$, conditioned on E_ε , we have

$$\tilde{\mu} - t_\varepsilon = \mu + \text{AN}_\mu - t_\varepsilon > \mu - 2t_\varepsilon > 2t_\varepsilon > 0,$$

and

$$\tilde{\mu} - t_\varepsilon = \mu + \text{AN}_\mu - t_\varepsilon < \mu.$$

Let $f(x) = 1/\sqrt{x}$. By the mean value theorem, we have

$$f(\tilde{\mu} - t_\varepsilon) - f(\mu) = f'(a) (\tilde{\mu} - t_\varepsilon - \mu),$$

for some a between μ and $\tilde{\mu} - t_\varepsilon$. Since $f'(x) = -\frac{1}{2x^{3/2}}$ and $x \mapsto |f'(x)|$ is decreasing for $x > 0$, we can bound

$$|f'(a)| \leq \frac{1}{2(\tilde{\mu} - t_\varepsilon)^{3/2}}$$

Thus,

$$\begin{aligned} \Delta_2 &= \sqrt{A} \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \left| \frac{1}{\sqrt{\tilde{\mu} - t_\varepsilon}} - \frac{1}{\sqrt{\mu}} \right| \\ &\leq \sqrt{A} \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \frac{|\tilde{\mu} - t_\varepsilon - \mu|}{2(\tilde{\mu} - t_\varepsilon)^{3/2}} \\ &\stackrel{(i)}{\leq} \sqrt{A} \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \frac{2t_\varepsilon}{2(\mu - 2t_\varepsilon)^{3/2}} \\ &\stackrel{(ii)}{\leq} \sqrt{A} \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \frac{2\sqrt{2}t_\varepsilon}{\mu^{3/2}} \\ &\stackrel{(iii)}{\leq} \sqrt{A} \mathbb{1}\{\hat{\mu} > 4t_\varepsilon\} \frac{\sqrt{2}t_\varepsilon}{\mu} \\ &\leq \sqrt{2At_\varepsilon} \frac{\mathbb{1}\{\mu > 0\}}{\mu} \end{aligned}$$

where each step uses

- (i) the noise concentration event E_ε
- (ii) $\mu > 4t_\varepsilon$ implies $\mu - 2t_\varepsilon > \mu/2$
- (iii) $\mu > 4t_\varepsilon$ implies $\sqrt{t_\varepsilon/\mu} < 1/2$

Using the same argument as in the proof of Lemma 19, we have

$$\mathbb{E}[\Delta_2] \lesssim \sqrt{At_\varepsilon} (\zeta n h)^{-1} h^{-(d+\alpha)}.$$

□

F.6 Proof of Lemma 20.

Proof. Recall that $\hat{\mu}$ is the empirical average of i.i.d. Bernoulli random variables, where the success probability $\sigma(\mathbf{x})$, for any context $\mathbf{x} \in \text{supp}(Q)$, is defined as the measure of the joint context-price bin under the training distribution μ :

$$\sigma(\mathbf{x}) = \mu(A_{h,\mathcal{X}}(\mathbf{x}) \times A_{h,\mathcal{T}}(t^\dagger(\mathbf{x}))).$$

We decompose the expectation by conditioning on whether $\sigma(\mathbf{X}^Q)$ is small or large.

$$\begin{aligned} \mathbb{E}[\mathbb{1}(\hat{\mu} < 4t_\varepsilon)] &= \mathbb{E}[\mathbb{1}(\hat{\mu} < 4t_\varepsilon, \sigma(\mathbf{X}^Q) > 8t_\varepsilon)] + \mathbb{E}[\mathbb{1}(\hat{\mu} < 4t_\varepsilon, \sigma(\mathbf{X}^Q) < 8t_\varepsilon)] \\ &\leq \mathbb{E}[\mathbb{1}(\hat{\mu} < 4t_\varepsilon, \sigma(\mathbf{X}^Q) > 8t_\varepsilon)] + \mathbb{E}[\mathbb{1}(\sigma(\mathbf{X}^Q) < 8t_\varepsilon)] \\ &=: E_1 + E_2. \end{aligned}$$

Bounding E_1 . In the regime where $\sigma(\mathbf{x}) \geq 8t_\varepsilon$, the threshold $4t_\varepsilon$ satisfies $4t_\varepsilon \leq \sigma(\mathbf{x})/2$. We apply the multiplicative Chernoff bound for the lower tail of a sum of n i.i.d. Bernoulli variables with mean $\sigma(\mathbf{x})$:

$$\mathbb{P}(\hat{\mu} < (1 - \delta)\sigma(\mathbf{x})) \leq \exp\left(-\frac{\delta^2 n \sigma(\mathbf{x})}{2}\right).$$

Setting $\delta = 1/2$, we have $(1 - \delta)\sigma(\mathbf{x}) = \sigma(\mathbf{x})/2 \geq 4t_\varepsilon$. Thus:

$$E_2 = \mathbb{E}\left[\mathbb{P}(\hat{\mu} < 4t_\varepsilon \mid \mathbf{X}^Q)\right] \leq \mathbb{E}\left[\exp\left(-\frac{n\sigma(\mathbf{X}^Q)}{8}\right)\right] \stackrel{(i)}{\lesssim} \mathbb{E}\left[\frac{1}{n\sigma(\mathbf{X}^Q)}\right] \stackrel{(ii)}{\lesssim} (\zeta n h)^{-1} h^{-(d+\alpha)}.$$

where each step uses:

- (i) $e^{-z} \leq \frac{1}{ez}$ for $z > 0$
- (ii) Expectation technique in used in Appendix E.4.1.

Bounding E_2 . We start by bounding $\sigma(\mathbf{X}^Q)$ from below:

$$\sigma(\mathbf{X}^Q) \stackrel{(i)}{\gtrsim} h\zeta P(A_{h,\mathcal{X}}(\mathbf{X}^Q)) \stackrel{(ii)}{\gtrsim} h^{\alpha+1}\zeta Q(A_{h,\mathcal{X}}(\mathbf{X}^Q))$$

where each step uses

- (i) Near-optimal price exploration assumption (Assumption ??),
- (ii) Transfer exponent assumption (Definition 1)

This lower bound on $\sigma(\mathbf{X}^Q)$ implies

$$E_2 = \mathbb{E}\left[\mathbb{1}(\sigma(\mathbf{X}^Q) < 8t_\varepsilon)\right] \lesssim \mathbb{E}\left[\mathbb{1}\left(h^{\alpha+1}\zeta Q(A_{h,\mathcal{X}}(\mathbf{X}^Q)) < 8t_\varepsilon\right)\right]. \quad (62)$$

The term E_2 represents the probability that a target context $\mathbf{X}^Q$ falls into a "rare bin". To ensure the stability of the DP mechanism, E_2 must vanish at a rate consistent with the primary privacy-variance term, specifically $E_2 \lesssim (\zeta n h)^{-1} h^{-(d+\alpha)}$. To this end, we need some regularity condition on Q ; Q should not have many thin regions. To exclude such pathological cases, we impose a Density Power Condition (or Lower Margin Condition) on Q . This regularity condition, which can be found in nonparametric minimax analysis literature [Rigollet, 2007; Audibert & Tsybakov, 2007], ensures that the measure of contexts with small surrounding bin mass is controlled.

Assumption 2 (Context Margin Condition). There exists a constant $C_M > 0$ such that for a density power $\gamma = 2$, the target context distribution Q satisfies the following condition for any threshold $t > 0$:

$$\mathbb{P}\left(Q(A_{h,\mathcal{X}}(\mathbf{X}^Q)) < t\right) \leq C_M t^\gamma, \quad (63)$$

where $A_{h,\mathcal{X}}(\mathbf{x})$ is the context bin of width h around $\mathbf{x}$.

Let $t^* := \frac{8t_\varepsilon}{\zeta h^{\alpha+1}}$. By Assumption 2 with $\gamma = 2$, we bound the probability of this event:

$$E_2 \leq \mathbb{P}\left(Q(A_{h,\mathcal{X}}(\mathbf{X}^Q)) < t^*\right) \leq C_M (t^*)^2 = C_M \left(\frac{8t_\varepsilon}{\zeta h^{\alpha+1}}\right)^2 \lesssim \frac{1}{n\varepsilon^2 \zeta^2 h^{2\alpha+2}} \quad (64)$$

Algorithm 1 LDP Off-Policy Pricing with Pessimistic Estimation

Require: Privatized data $\tilde{\mathcal{D}}_n = \{(\tilde{\mathbf{U}}_i, \tilde{\mathbf{V}}_i)\}_{i=1}^n$, target context $\mathbf{x}^Q$, bandwidth h , privacy budget ε

- 1: Set noise variance $\sigma_\varepsilon^2 \leftarrow 64/\varepsilon^2$
- 2: Set stability threshold $\tau_{n,\varepsilon} \leftarrow \sqrt{n\sigma_\varepsilon^2 \log n}$
- 3: Compute candidate bin set

$$\mathcal{J}_{\mathbf{x}^Q} \leftarrow \{j \in \{1, \dots, K\} : B_j \cap (\{\mathbf{x}^Q\} \times \mathcal{T}) \neq \emptyset\}$$

- 4: Initialize $\text{LCB}_{\max} \leftarrow -\infty$
- 5: Initialize $\hat{\pi}_h^\varepsilon(\mathbf{x}^Q)$ arbitrarily in $\mathcal{T}$
- 6: **for** $j \in \mathcal{J}_{\mathbf{x}^Q}$ **do**
- 7: Let t_j be the centroid of the treatment interval of bin j
- 8: Compute privatized counts:

$$\tilde{N} \leftarrow \sum_{i=1}^n \tilde{U}_{i,j}, \quad \tilde{S} \leftarrow \sum_{i=1}^n \tilde{V}_{i,j}$$

- 9: **if** $\tilde{N} < \tau_{n,\varepsilon}$ **then**
- 10: $\text{LCB}_j \leftarrow -\infty$ $\triangleright$ Unstable bin
- 11: **else**
- 12: Compute confidence width:

$$\text{Width}_j \leftarrow C_{\text{LDP}} \frac{\sqrt{\max\{n\sigma_\varepsilon^2, \tilde{N}\} \log n}}{\tilde{N}} + C_{\text{H}} h^\beta$$

- 13: $\text{LCB}_j \leftarrow (\tilde{S}/\tilde{N}) - \text{Width}_j$
 - 14: **if** $\text{LCB}_j > \text{LCB}_{\max}$ **then**
 - 15: $\text{LCB}_{\max} \leftarrow \text{LCB}_j$
 - 16: $\hat{\pi}_h^\varepsilon(\mathbf{x}^Q) \leftarrow t_j$
 - 17: **return** $\hat{\pi}_h^\varepsilon(\mathbf{x}^Q)$
-

where last inequality uses $t_\varepsilon \asymp (n\varepsilon^2)^{-1/2}$.

The target sub-optimal gap order is $(\zeta nh)^{-1} h^{-(d+\alpha)} = \frac{1}{\zeta n h^{d+\alpha+1}}$. We check the dominance of the powers of h : Under the geometric assumption $d > \alpha + 1$, we define the positive constant $\delta = d - \alpha - 1$. Comparing the rare-event sub-optimal gap E_1 to the privacy-induced variance T_3 , we obtain:

$$\frac{E_1}{E_2} \asymp \frac{\varepsilon^2 h^\delta}{\zeta}. \quad (65)$$

□